
2026 Spring Moduli Space Note

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1 2026.3.11

1.1 Presheaves and Sheaves

Definition 1.1. A **site** $(\mathcal{C}, \text{Cov}(\mathcal{C}))$ is a category $\mathcal{C}$ together with a Grothendieck topology $\text{Cov}(\mathcal{C})$.

Example 1.2. There are several foundational examples of sites used in algebraic geometry:

1. **Small étale site:** Denoted by $X_{\text{sm.ét}} = (\mathcal{C}, \text{Cov}(\mathcal{C}))$. For a scheme X , the underlying category is $\mathcal{C} = \{Y \rightarrow X \text{ étale}\}$, and $\text{Cov}(\mathcal{C})$ is the étale topology. This site is primarily used in the study of the homology and cohomology of X .
2. **Big étale site:** Denoted by $X_{\text{ét}} = (\mathcal{C}', \text{Cov}(\mathcal{C}'))$. For a scheme X , the category is defined as $\mathcal{C}' = \{Y \rightarrow X \text{ any morphism}\}$, and $\text{Cov}(\mathcal{C}')$ is the étale topology. A covering is given by a jointly surjective family of morphisms $\{U_i \rightarrow U\}$. The big étale site is extensively used in moduli theory.
3. **Smooth site.**
4. **fppf and fpqc sites.**

Note that the smooth, fppf, and fpqc sites are essential foundations for both moduli theory and stack theory.

Motivation

The primary motivation for introducing these structures is to study the properties of moduli functors. A moduli functor is initially considered as a presheaf. To study it effectively, we need to introduce the notion of sheaves in this generalized setting, which essentially necessitates the language of sites.

- Recall the classical case of a topological space X . A presheaf on X is a contravariant functor $\mathcal{F} : \text{Open}(X)^{\text{op}} \rightarrow \text{Set}$. The presheaf $\mathcal{F}$ is a sheaf if it satisfies two fundamental axioms: the identity axiom and the gluing axiom.

In modern algebraic geometry, we need to generalize this classical topological notion of sheaves, and this is achieved precisely by utilizing the language of sites.

- If a moduli space is a fine moduli space, then the representable functor $\mathcal{M} = \text{Hom}(-, M)$ is a sheaf.

Slogan: Define the moduli functor $\rightarrow$ Hope it is representable $\rightarrow$ Verify the sheaf conditions $\rightarrow$ Realize that non-trivial automorphisms break the sheaf axioms $\rightarrow$ Introduce stacks to keep track of the automorphisms.

Definition 1.3 (Presheaf and Sheaf). Let $\mathcal{C}$ be a category.

1. A **presheaf** of sets on $\mathcal{C}$ is a contravariant functor $\mathcal{M} : \mathcal{C}^{\text{op}} \rightarrow (\text{Sets})$.
2. Given a Grothendieck topology $\text{Cov}(\mathcal{C})$ on $\mathcal{C}$, a presheaf $\mathcal{M}$ is called a **sheaf** over the site $(\mathcal{C}, \text{Cov}(\mathcal{C}))$ if for every covering family $\{U_i \rightarrow U\} \in \text{Cov}(\mathcal{C})$, the following diagram is an equalizer:

$$\mathcal{M}(U) \longrightarrow \prod_i \mathcal{M}(U_i) \rightrightarrows \prod_{i,j} \mathcal{M}(U_i \times_U U_j)$$

This equalizer condition perfectly encapsulates both the identity and gluing axioms.

An important result in this context is the following theorem concerning the existence of sheafification.

Theorem 1.4 (Existence of Sheafification). For every presheaf $\mathcal{M}$ over a site $(\mathcal{C}, \text{Cov}(\mathcal{C}))$, there exists a canonical map $\mathcal{M} \rightarrow \mathcal{M}^*$ satisfying a universal property, where $\mathcal{M}^*$ is a sheaf.

Idea. Using a construction similar to that found in standard sheaf theory, one can first obtain a canonical map $\mathcal{M} \rightarrow \mathcal{M}^+$ such that $\mathcal{M}^+$ is a separated presheaf.

Here, a presheaf is said to be **separated** if it satisfies the identity axiom. That is, for all sections $s, t \in \mathcal{M}(U)$ such that their restrictions agree on the cover, meaning $s|_{U_i} = t|_{U_i}$ for all i , it necessarily follows that $s = t$.

The map $\mathcal{M} \rightarrow \mathcal{M}^{++}$ is a sheafification, where

$$\mathcal{M}^+(U) = \varinjlim_{\mathcal{U}} \mathcal{M}(\mathcal{U})$$

and

$$\mathcal{M}(\mathcal{U}) = \left\{ (s_i)_{i \in I} \in \prod_{i \in I} \mathcal{M}(U_i) \mid s_i|_{U_i \times_U U_j} = s_j|_{U_i \times_U U_j} \right\},$$

for a covering $\mathcal{U} = \{U_i \rightarrow U\}_{i \in I}$. □

Remark. This raises a natural question: is every representable functor $\text{Hom}(-, X)$ a sheaf?

Definition 1.5 (Canonical Topology). The **canonical topology** on a category $\mathcal{C}$ is the finest Grothendieck topology such that all representable functors are sheaves.

Example 1.6. We have the following examples of topologies regarding representable functors:

1. In the Zariski topology, representable functors are indeed sheaves.
2. For the fpqc topology, representable functors are also sheaves. It is likely the largest topology with this property; for strictly larger topologies, counterexamples exist.

These concepts have deep connections with Descent Theory.

1.2 Prestacks, CFG and Stacks

Motivation

Consider the moduli functor $\mathcal{M}_g: (\text{Sch})^{\text{op}} \longrightarrow (\text{Sets})$ defined by mapping a scheme S to the set of isomorphism classes of curves over S :

$$S \longmapsto \{\mathcal{X} \rightarrow S\} / \cong$$

This functor exhibits the following issues:

1. It is not a sheaf, even in the Zariski topology.
2. It is not representable.

This failure occurs because there exist non-trivial automorphisms. The essence of the problem is that locally on a cover $\{U_i\}$, we might have two families $\mathcal{X}|_{U_i} \rightarrow S$ and $\mathcal{X}'|_{U_i} \rightarrow S$ which are identically isomorphic in $\mathcal{M}_g(U_i)$, but these do not necessarily yield the same global structure in $\mathcal{M}_g(X)$. This inability to glue uniquely is precisely due to the presence of automorphisms, which cause the local pieces to twist.

A better approach is to define a new category $\mathcal{M}_g$. The objects of this category are pairs (U, C) , where U is a scheme (i.e., $U \in (\text{Sch})$) and $C \rightarrow U$ is a family of smooth curves of genus g .

A morphism from (V, D) to (U, C) is given by a commutative Cartesian diagram:

$$\begin{array}{ccc} D & \longrightarrow & C \\ \downarrow & & \downarrow \\ V & \longrightarrow & U \end{array}$$

where $D \rightarrow C \times_U V$ is an isomorphism.

There exists a natural functor $\pi: \mathcal{M}_g \rightarrow (\text{Sch})$ defined by mapping the object (U, C) to U . For every scheme $U \in (\text{Sch})$, the fiber $\pi^{-1}(U)$ corresponds exactly to $\mathcal{M}_g(U)$ with isomorphism information. Formally, it forms the category of smooth curves of genus g over U , complete with isomorphisms.

Remark. *The two constructions of $\mathcal{M}_g$ are exactly different.*

Definition 1.7 (Groupoid). A **groupoid**, which can be viewed as a generalization of a group, is a category in which every morphism is an isomorphism.

Definition 1.8. A functor $\pi: \mathcal{S} \rightarrow \mathcal{C}$ is said to be **fibered in groupoids over the base category $\mathcal{C}$** if it satisfies the following two conditions:

1. For every morphism $f: V \rightarrow U$ in the category $\mathcal{C}$, and for every object $x \in \mathcal{S}$ such that $\pi(x) = U$, there exists a lift $\phi: y \rightarrow x$ in $\mathcal{S}$ such that $\pi(\phi) = f$. (This behaves conceptually similarly to a forgetful functor).
2. (Uniqueness of lifts). For any morphisms $\phi: y \rightarrow x$ and $\psi: z \rightarrow x$ in the category $\mathcal{S}$, and any morphism $f: \pi(z) \rightarrow \pi(y)$ in $\mathcal{C}$ such that $\pi(\phi) \circ f = \pi(\psi)$, there exists a unique lift $\chi: z \rightarrow y$ in $\mathcal{S}$ such that $\phi \circ \chi = \psi$ and $\pi(\chi) = f$.

Example 1.9. Consider the moduli category $\mathcal{M}_g$. For any morphism $V \rightarrow U$ and any family of curves $C \rightarrow U$ in $\mathcal{M}_g(U)$, the lifting is obtained by forming a Cartesian diagram:

$$\begin{array}{ccc} C \times_U V & \longrightarrow & C \\ \downarrow & \square & \downarrow \\ V & \longrightarrow & U \end{array}$$

Definition 1.10 (Category Fibered in Groupoids). A functor $\pi: \mathcal{S} \rightarrow \mathcal{C}$ is a **category fibered in groupoids** (CFG) if it is fibered in groupoids over $\mathcal{C}$ and the fiber category $\pi^{-1}(U)$ is a groupoid for any object $U \in \mathcal{C}$.

Definition 1.11 (Prestack). A category fibered in groupoids $\pi: \mathcal{S} \rightarrow \mathcal{C}$ is called a **prestack** if for all objects $U \in \mathcal{C}$ and all objects $x, y \in \pi^{-1}(U)$, the presheaf on the slice category $(\mathcal{C}/U)$,

$$\underline{\text{Isom}}_{\mathcal{S}}(x, y): (\mathcal{C}/U)^{\text{op}} \rightarrow \text{Sets}$$

defined by assigning to each morphism $(f: V \rightarrow U)$ the set of isomorphisms $\text{Isom}_{\pi^{-1}(V)}(f^*x, f^*y)$, is a sheaf.

Remark. By the definition of fibered in groupoids over $\mathcal{C}$, f^*x and f^*y are well-defined objects in $\pi^{-1}(V)$ up to canonical isomorphism.

Example 1.12. The natural functor $\pi: \mathcal{M}_g \rightarrow (\text{Sch})$ is a prestack.

For any two families of curves $C_1 \rightarrow U$ and $C_2 \rightarrow U$ over U , and for any morphism $V \rightarrow U$, the presheaf assigns this morphism to the set of isomorphisms between the pullbacks:

$$\text{Isom}(\tilde{C}_1 \rightarrow V, \tilde{C}_2 \rightarrow V)$$

where $\tilde{C}_1 = C_1 \times_U V$ and $\tilde{C}_2 = C_2 \times_U V$. In fact, verifying that this presheaf satisfies the sheaf condition is a trivial exercise.

Question: How can we consider the category of schemes (Sch) as a prestack?

Answer: We first consider a scheme X via its functor of points, identifying it with the representable functor $\text{Hom}(-, X)$. Recall the moduli functor $\mathcal{M}_g$. If $\mathcal{M}_g$ is representable by a fine moduli space M_g , then we have the following identification for the fibers:

$$\mathcal{M}_g(U) = \pi^{-1}(U) = \text{Hom}(U, M_g).$$

This insight naturally leads us to consider a category fibered in groupoids $\pi: \mathcal{S} \rightarrow (\text{Sch})$, where the fiber is given by $\pi^{-1}(U) = \text{Hom}(U, X)$ for some fixed scheme X .

In this setup, $\mathcal{S}$ represents the pairs of the form $(U \rightarrow X)$. Indeed, the objects of $\mathcal{S}$ are exactly the objects of the slice category (Sch/X) . The formulation of the morphisms in this category is left as an exercise for the reader to think about.

Definition 1.13 (Stack). Let $\mathcal{C}$ be a site. A prestack $\pi: \mathcal{S} \rightarrow \mathcal{C}$ is called a **stack** if it satisfies the following descent condition.

For every covering family $\{U_i \rightarrow U\}_{i \in I}$ in $\text{Cov}(\mathcal{C})$, and for any collection of local objects $x_i \in \pi^{-1}(U_i)$ along with isomorphisms

$$\phi_{ij}: x_i|_{U_i \times_U U_j} \xrightarrow{\sim} x_j|_{U_i \times_U U_j}$$

satisfying the cocycle condition $\phi_{ik} = \phi_{jk} \circ \phi_{ij}$ on the triple intersections $U_i \times_U U_j \times_U U_k$, there exists a global object $x \in \pi^{-1}(U)$ and isomorphisms $\psi_i: x|_{U_i} \xrightarrow{\sim} x_i$ such that the

compatibility condition

$$\phi_{ij} = (\psi_j|_{U_i \times_U U_j}) \circ (\psi_i|_{U_i \times_U U_j})^{-1}$$

holds on the double intersections $U_i \times_U U_j$.

Remark. *In the definition of descent data, we employ the restriction notation for brevity. Strictly speaking, the "restriction" $x_i|_{U_i \times_U U_j}$ denotes the pullback $p_1^*x_i$ along the first projection $p_1 : U_i \times_U U_j \rightarrow U_i$. Similarly, $x_j|_{U_i \times_U U_j}$ denotes $p_2^*x_j$. Furthermore, the cocycle condition strictly holds in the fiber category $\pi^{-1}(U_i \times_U U_j \times_U U_k)$ after pulling back the isomorphisms $\phi_{\bullet\bullet}$ via the respective canonical projections from the triple intersection.*

2 2026.3.16

Review from last week:

There is a direct analogy between the conditions required for a category fibered in groupoids to be a stack, and the conditions for a presheaf to be a sheaf. We can summarize this as follows:

$$\begin{array}{ccccc}
 \text{Prestack} & & + & \text{descent condition} & = & \text{Stack} \\
 \downarrow & & & \downarrow & & \downarrow \\
 \text{Presheaf} + \text{separated} & & & \text{gluing axiom} & & \text{Sheaf} \\
 (\text{identity axiom}) & & & & &
 \end{array}$$

2.1 Relation to Moduli Functors

We can define the following mappings between CFG and moduli functors:

$$\text{CFG } \mathcal{S} \xrightarrow{\pi} \mathcal{C} \rightleftharpoons \text{Moduli functors } \mathcal{M} : \mathcal{C}^{\text{op}} \longrightarrow (\text{Sets})$$

From left to right

Given a category fibered in groupoids $\mathcal{S} \xrightarrow{\pi} \mathcal{C}$, we define a moduli functor $\mathcal{M}_{\pi} : \mathcal{C}^{\text{op}} \longrightarrow (\text{Sets})$ by mapping each object to the set of isomorphism classes of its fiber:

$$B \longmapsto \pi^{-1}(B) / \cong$$

From right to left

Given a moduli functor $\mathcal{M} : \mathcal{C}^{\text{op}} \longrightarrow (\text{Sets})$, we can define a category fibered in groupoids $\mathcal{S}$ as follows:

- The objects of $\mathcal{S}$ are defined as pairs of an object in $\mathcal{C}$ and an element of its moduli set:

$$\text{obj}(\mathcal{S}) := \{(U, \zeta) \mid U \in \text{obj}(\mathcal{C}), \zeta \in \mathcal{M}(U)\}$$

- For any objects (U, ζ) and (V, η) in $\mathcal{S}$, the morphisms from (V, η) to (U, ζ) are defined as the morphisms in the base category that pull back the moduli element appropriately:

$$\text{Hom}_{\mathcal{S}}((V, \eta), (U, \zeta)) := \{f : V \rightarrow U \text{ in } \mathcal{C} \text{ such that } \mathcal{M}(f)(\zeta) = \eta\}$$

Remark. *This is not necessarily equivalence.*

Proposition 2.1. Suppose $\mathcal{S} \xrightarrow{\pi} \mathcal{C}$ is a category fibered in sets. Then we have the following properties:

1. $\mathcal{S} \xrightarrow{\pi} \mathcal{C}$ is a prestack if and only if the corresponding functor $\mathcal{M}_\pi$ is separated.
2. $\mathcal{S} \xrightarrow{\pi} \mathcal{C}$ is a stack if and only if the corresponding functor $\mathcal{M}_\pi$ is a sheaf.

2.2 Examples of stacks

Example 2.2 (Sheaves as stacks). Let $\mathcal{F}$ be a sheaf of sets on a site $\mathcal{C} = (\mathcal{C}, \text{Cov}(\mathcal{C}))$. We can define a category fibered in groupoids $\mathcal{S} \xrightarrow{\pi} \mathcal{C}$ in the following way:

- **Objects:** The objects are pairs (U, s) such that $U \in \text{obj}(\mathcal{C})$ and $s \in \mathcal{F}(U)$.
- **Morphisms:** A morphism from (V, t) to (U, s) is given by a morphism $f : V \rightarrow U$ in $\mathcal{C}$ such that $f^*s = t$ (where f^* denotes the restriction map associated to the sheaf $\mathcal{F}$).

The functor $\pi : \mathcal{S} \rightarrow \mathcal{C}$ is the forgetful functor. The fiber $\pi^{-1}(U)$ is a discrete groupoid (as a set), meaning that the only isomorphisms are the identity morphisms. One can verify this.

Then $\pi : \mathcal{S} \rightarrow \mathcal{C}$ is a stack.

Proof. Why is $\pi : \mathcal{S} \rightarrow \mathcal{C}$ a CFG?

To prove that $\pi : \mathcal{S} \rightarrow \mathcal{C}$ is a CFG, we need to prove the existence and uniqueness of liftings.

Existence of lifting: For all morphisms $f : V \rightarrow U$ in $\mathcal{C}$ and an object (U, s) over U , we need to find an object (V, t) and a morphism $\Phi : (V, t) \rightarrow (U, s)$ such that its corresponding map in the base category is f . We simply take $t = f^*s$.

Uniqueness of lifting: Suppose we have morphisms $\Phi : (V, t) \rightarrow (U, s)$ and $\Psi : (W, w) \rightarrow (U, s)$ in $\mathcal{S}$, and a morphism $h : W \rightarrow V$ in $\mathcal{C}$ such that $\pi(\Phi) \circ h = \pi(\Psi)$. We need to find a unique morphism $\chi : (W, w) \rightarrow (V, t)$ with $\pi(\chi) = h$ such that $\Phi \circ \chi = \Psi$. Here, we just take the natural lift of h given by the “existence of lifting” part proven above.

Why is $\pi : \mathcal{S} \rightarrow \mathcal{C}$ a prestack?

To show that $\pi : \mathcal{S} \rightarrow \mathcal{C}$ is a prestack, we need to check that the Isom functor is a sheaf. We check the functor $\text{Isom}((U, s), (U, t)) : (\mathcal{C}/U)^{\text{op}} \rightarrow (\text{Sets})$. Let us denote this functor

by I . It maps an object $(V \xrightarrow{f} U)$ to the set $I(V \xrightarrow{f} U)$, which is defined as:

$$I(V \xrightarrow{f} U) = \begin{cases} \{\text{id}\} & \text{if } f^*s = f^*t, \\ \emptyset & \text{if not.} \end{cases}$$

One can verify that this functor I is a sheaf.

Why is $\pi : \mathcal{S} \rightarrow \mathcal{C}$ a stack?

To show that $\pi : \mathcal{S} \rightarrow \mathcal{C}$ is a stack, we need to check the descent condition.

For any covering $\{U_i \rightarrow U\}$ in $\mathcal{C}$, let $x_i \in \pi^{-1}(U_i)$ and let

$$\phi_{ij} : x_i|_{U_i \times_U U_j} \xrightarrow{\sim} x_j|_{U_i \times_U U_j}$$

be the descent data. Here, the objects are of the form $x_i = (U_i, s_i)$ where $s_i \in \mathcal{F}(U_i)$.

The condition that ϕ_{ij} is an isomorphism is equivalent to saying that $s_i|_{U_{ij}} = s_j|_{U_{ij}}$ in $\mathcal{F}(U_{ij})$. (In this discrete case, the concept of “isomorphism” reduces to simple equality).

As $\mathcal{F}$ is a sheaf, the matching sections s_i glue together. Therefore, there exists a unique section $s \in \mathcal{F}(U)$ such that $s_i = s|_{U_i}$ for all i .

We can take $x = (U, s)$ with $s \in \mathcal{F}(U)$ as the desired object in $\pi^{-1}(U)$, which proves the descent condition. $\square$

Example 2.3 (The moduli space $\mathcal{M}_g$). Consider the category fibered in groupoids $\mathcal{M}_g \xrightarrow{\pi} (\text{Scheme})$.

- **Objects:** The objects are pairs $(U, C \rightarrow U)$, where $C \rightarrow U$ is a family of smooth projective curves of genus g .
- **Morphisms:** The morphisms are given by pullbacks.

Then $\pi : \mathcal{M}_g \rightarrow (\text{Scheme})$ is a stack.

Proof. We skip the proof that this is a CFG.

Why is $\pi : \mathcal{M}_g \rightarrow (\text{Scheme})$ a prestack?

To check if it is a prestack, let $C_1, C_2 \in \mathcal{M}_g(U) = \pi^{-1}(U)$. We need to verify that the presheaf

$$\text{Isom}(C_1, C_2) : \text{Sch}/U \rightarrow \text{Sets}$$

defined by

$$(V \rightarrow U) \mapsto \text{Isom}(C_1 \times_U V, C_2 \times_U V)$$

is a sheaf.

That is, for any open covering $\{U_i \rightarrow U\}$, if we have isomorphisms

$$C_1 \times_U U_i \xrightarrow{\cong} C_2 \times_U U_i$$

for all i that are compatible on the overlaps, then they glue to give an isomorphism $C_1 \cong C_2$. This is trivial in the Zariski topology.

Remark. *If we consider other sites such as the étale, smooth, or fpqc sites, this property still holds. One can easily reduce to the case where the covering consists of a single morphism, i.e., $\{U_i \rightarrow U\} = \{V \rightarrow U\}$ (only one index), by taking $V = \coprod U_i \rightarrow U$.*

If there is an isomorphism $\phi : C_1 \times_U V \xrightarrow{\sim} C_2 \times_U V$ over V such that its two pullbacks to $V \times_U V$ agree (i.e., it satisfies the cocycle condition $p_1^ \phi = p_2^* \phi$), then ϕ uniquely descends to an isomorphism $C_1 \cong C_2$ over U .*

In fact, this relies on the covering morphism being faithfully flat. By the descent theorem in commutative algebra, we obtain the existence of gluing isomorphisms.

Why is $\pi : \mathcal{M}_g \rightarrow (\mathbf{Scheme})$ a stack?

To check the descent condition, let $\{U_i \rightarrow U\}$ be a covering. Suppose we are given objects $\pi_i : C_i \rightarrow U_i$ in $\mathcal{M}_g(U_i)$ and isomorphisms

$$\phi_{ij} : C_i|_{U_{ij}} \xrightarrow{\cong} C_j|_{U_{ij}}$$

satisfying the standard cocycle conditions.

The question is whether there exists a family $C \rightarrow U$ such that its restriction $C|_{U_i}$ is isomorphic to $C_i \rightarrow U_i$. The scheme C is constructed by gluing the schemes C_i along the isomorphisms ϕ_{ij} (this gluing process is trivial in the Zariski topology). Thus, effective descent holds. $\square$

2.3 Quotient stacks

Definition 2.4 (Quotient stack). Let X be a scheme and G be an algebraic group acting properly on X . We define the **quotient stack** $\pi : [X/G] \rightarrow (\mathbf{Sch})$ as follows:

- **Objects:** An object over a scheme U is a pair $(P \rightarrow U, f)$, where $P \rightarrow U$ is a principal G -bundle and $f : P \rightarrow X$ is a G -equivariant morphism. This can be

summarized by the following diagram:

$$\begin{array}{ccc} P & \xrightarrow{f} & X \\ \downarrow \text{\scriptsize G-bundle} & & \\ U & & \end{array}$$

Remark (Intuitive understanding). *Geometrically, one can visualize the objects as families of G -torsors over the base space U , equipped with a compatible map to the space X where the group action takes place.*

Remark. $[X/G](U) := \pi^{-1}(U)$.

Let us evaluate the stack at a single point to see its relation to the classical quotient. The objects of $[X/G](\text{pt})$ consist of a principal G -bundle over a point (which is trivially isomorphic to G itself) and a G -equivariant map $f : G \rightarrow X$. The set of isomorphism classes of such objects corresponds precisely to the space of G -orbits:

$$\text{obj}([X/G](\text{pt})) = \{(\text{pt}, G \xrightarrow{f} X)\} = X$$

However, the morphisms between two objects $x, y \in X$ is given by the set of group elements $g \in G$ such that $g \cdot x = y$.

Example 2.5. The moduli space of curves $\mathcal{M}_g$ can be constructed and understood as a quotient stack:

$$\mathcal{M}_g = [U/\text{PGL}(N)]$$

where $U \subseteq \text{Hilb}_{P(z), \mathbb{P}^N}$ is a suitable open subset of the Hilbert scheme, and $P(z)$ denotes the corresponding Hilbert polynomial.

2.4 Stack of Vector Bundles

Consider the CFG $\text{Vect}_r \longrightarrow (\text{Sch})$ defined as follows:

- **Objects:** The objects are pairs $(U, \mathcal{E})$, where U is a scheme and $\mathcal{E}$ is a locally free sheaf on U of rank r .
- **Morphisms:** The morphisms are given by pullbacks.

Fact: This is actually a stack!

Remark. *Although there are very few constraints on the objects, it still forms a stack, which is quite remarkable.*

Given a curve C , let $N_r(C)$ be the moduli space of vector bundles of rank r on C . This corresponds to the set of isomorphism classes of vector bundles of rank r on C .

However, this space is not well-behaved because it is not of finite type. For example, consider the case where $C = \mathbb{P}^1$. The space $N_r(\mathbb{P}^1)$ consists of infinitely many discrete points. By Grothendieck's theorem, every vector bundle on $\mathbb{P}^1$ splits as a direct sum of line bundles of the form $\bigoplus \mathcal{O}(m_i)$, and there are infinitely many choices for the integers m_i .

To resolve this issue, we restrict our attention to vector bundles of a fixed degree. We define:

$$N_{r,d}(C) := \{\mathcal{E} \mid \text{rank}(\mathcal{E}) = r, \deg(\mathcal{E}) = d\}$$

This restricted moduli space is much better behaved and can be constructed as a projective scheme. (Note: to strictly obtain a projective scheme, one still needs to add some other conditions).

2.5 Stackification

Just as one can sheafify a presheaf, one can stackify a prestack.

Theorem 2.6 (Stackification). For any category fibered in groupoids (or prestack) $\mathcal{S} \rightarrow \mathcal{C}$, there exists a canonical stackification $\mathcal{S}^* \rightarrow \mathcal{C}$.

Remark. *This process is completely analogous to the sheafification of the associated moduli functor.*

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3.1 Representable Morphisms

Definition 3.1 (Representable CFG(Prestack)). A CFG(prestack) $\pi : \mathcal{S} \rightarrow \mathcal{C}$ is **representable** by an object $B \in \text{obj}(\mathcal{C})$ if π is given by the functor $\text{Mor}_{\mathcal{C}}(-, B) : \mathcal{C}^{\text{op}} \rightarrow (\text{Set})$ and by the mapping in 2.1. In fact in this case you can consider $\mathcal{S}$ as $\mathcal{C}/B$.

Example 3.2. Let X be a scheme. This naturally gives rise to a stack $\mathcal{S}_X \rightarrow (\text{Sch})$ where:

- The objects are $\text{obj}(\mathcal{S}_X) = \{U \rightarrow X\}$.
- The morphisms $\text{morph}(\mathcal{S}_X)$ are given by commutative triangles over X :

$$\begin{array}{ccc} U & & \\ \downarrow & \searrow & \\ V & \longrightarrow & X \end{array}$$

Definition 3.3 (2-Fibered Product). Let $\mathcal{X} \xrightarrow{p} \mathcal{Z}$ and $\mathcal{Y} \xrightarrow{q} \mathcal{Z}$ be morphisms of CFGs over $\mathcal{C}$. The **2-fibered product** is a CFG $\mathcal{X} \times_{\mathcal{Z}} \mathcal{Y} \rightarrow \mathcal{C}$ defined by the following data:

- Objects: Tuples (U, x, y, α) , where $U \in \text{obj}(\mathcal{C})$, $x \in \mathcal{X}(U)$, $y \in \mathcal{Y}(U)$, and α is an isomorphism $\alpha : p(x) \xrightarrow{\sim} q(y)$ in $\mathcal{Z}(U)$.
- Morphisms: A morphism is defined by a compatible diagram between these objects.

Definition 3.4 (Representable Morphism). Let $\mathcal{X}$ and $\mathcal{Y}$ be CFGs over a base category $\mathcal{C}$. A morphism of CFGs $f : \mathcal{X} \rightarrow \mathcal{Y}$ over $\mathcal{C}$ is called **representable** if for every object $U \in \text{obj}(\mathcal{C})$ and every morphism $(\mathcal{C}/U) \rightarrow \mathcal{Y}$, the 2-fibered product $\mathcal{X} \times_{\mathcal{Y}} (\mathcal{C}/U) \rightarrow \mathcal{C}$ is representable.

Remark. As we will see later, for algebraic stacks (where the base category is typically $\mathcal{C} = (\text{Sch})_{\text{ét}}$), we can introduce various notions of morphisms via the following Cartesian diagram:

$$\begin{array}{ccc} \mathcal{X} \times_{\mathcal{Y}} (\mathcal{C}/U) & \xrightarrow{f_U} & (\mathcal{C}/U) \\ \downarrow & & \downarrow \\ \mathcal{X} & \xrightarrow{f} & \mathcal{Y} \end{array}$$

That is, we say that a **representable morphism** f has a **property** P if the base change f_U

has property P for all morphisms $(\mathcal{C}/U) \rightarrow \mathcal{Y}$ from an object $U \in \text{obj}(\mathcal{C})$.

Lemma 3.5 (2-Yoneda Lemma). Let $\mathcal{S} \xrightarrow{\pi} \mathcal{C}$ be a CFG, and let U be an object in $\mathcal{C}$. Then there exists an equivalence of categories

$$\Phi : \text{Mor}(\mathcal{C}/U, \mathcal{S}) \xrightarrow{\sim} \pi^{-1}(U) = \mathcal{S}_U,$$

This equivalence is given by evaluating a morphism f at the identity morphism of U , that is,

$$\Phi(f) = f(\text{id}_U).$$

Remark. Consider a morphism $(\mathcal{C}/U) \rightarrow \mathcal{S} \times_{\mathcal{C}} \mathcal{S}$. This naturally gives rise to a pair of morphisms $X : (\mathcal{C}/U) \rightarrow \mathcal{S}$ and $Y : (\mathcal{C}/U) \rightarrow \mathcal{S}$. Consequently, by applying the 2-Yoneda Lemma, these morphisms correspond precisely to objects $X \in \text{obj}(\mathcal{S}_U)$ and $Y \in \text{obj}(\mathcal{S}_U)$ within the fiber category over U .

Example 3.6 (Representable Morphisms: The Diagonal). As a fundamental example of representable morphisms, we consider the diagonal morphism. For any CFG $\mathcal{S} \rightarrow \mathcal{C}$, there is a natural diagonal morphism of CFGs over $\mathcal{C}$:

$$\Delta : \mathcal{S} \longrightarrow \mathcal{S} \times_{\mathcal{C}} \mathcal{S}.$$

For all morphism of CFGs $(\mathcal{C}/U) \rightarrow \mathcal{S} \times_{\mathcal{C}} \mathcal{S}$ over the base category $\mathcal{C}$ where $U \in \text{obj}(\mathcal{C})$, by the 2-Yoneda Lemma (3.5), such a morphism corresponds to choosing a pair of objects (X, Y) in the fiber category $\mathcal{S}_U$. Recall that in the definition (1.11) of prestack, we defined:

$$\begin{aligned} \underline{\text{Isom}}_{\mathcal{S}}(X, Y) : (\mathcal{C}/U)^{\text{op}} &\longrightarrow \text{Sets} \\ (V \xrightarrow{f} U) &\longmapsto \text{Isom}_{\mathcal{S}_V}(f^*X, f^*Y) \end{aligned}$$

which by 2.1 defines a morphism of CFGs $\underline{\text{Isom}}_{\mathcal{S}}(X, Y) \rightarrow (\mathcal{C}/U)$.

Proposition 3.7. There is a Cartesian diagram of CFGs over $\mathcal{C}$:

$$\begin{array}{ccc} \underline{\text{Isom}}_{\mathcal{S}}(X, Y) & \longrightarrow & (\mathcal{C}/U) \\ \downarrow & & \downarrow \\ \mathcal{S} & \xrightarrow{\Delta} & \mathcal{S} \times_{\mathcal{C}} \mathcal{S} \end{array}$$

i.e. we can consider the CFG $\underline{\text{Isom}}_{\mathcal{S}}(X, Y)$ as the 2-fibered product $\mathcal{S} \times_{\mathcal{S} \times_{\mathcal{C}} \mathcal{S}} (\mathcal{C}/U)$.

Remark. If $\mathcal{S} \rightarrow \mathcal{C}$ is a prestack, then by the definition, for any two objects $X, Y \in \mathcal{S}_U$, the functor $\underline{\text{Isom}}_{\mathcal{S}}(X, Y) : (\mathcal{C}/U)^{op} \rightarrow \text{Sets}$ is guaranteed to be a sheaf. However, it is not necessarily representable by an object in the category $\mathcal{C}/U$.

Proposition 3.8. 1. The diagonal morphism Δ is representable if and only if the functor $\underline{\text{Isom}}_{\mathcal{S}}(X, Y)$ is representable for all pairs of objects (X, Y) .

2. Assume that the underlying site is subcanonical (that is, every representable functor is a sheaf). Under this assumption, if the diagonal Δ is representable, then it implies that the CFG $\mathcal{S} \rightarrow \mathcal{C}$ is a prestack.

3.2 Algebraic Spaces

Remark. Consider a CFG $\mathcal{S} \rightarrow (\text{Sch})$ over the category of schemes endowed with various Grothendieck topologies such as the Zariski, étale, smooth, or fppf topologies. Suppose $\mathcal{S}$ is a prestack with respect to one of these choices.

Here is an equivalence: $\mathcal{S} \rightarrow (\text{Sch})_{\text{zar}}$ is a stack with respect to the Zariski topology, and $\mathcal{S}$ satisfies the descent axiom along affine étale morphisms (meaning that compatible affine local data implies there exists a global section), if and only if $\mathcal{S} \rightarrow (\text{Sch})_{\text{ét}}$ is a stack with respect to the étale topology.

It implies that the Zariski is much coarser than the étale topology. That's a reason why we will mostly work with the étale topology in the context of algebraic stacks.

Remark. Henceforth we have two simplifications of notations:

- For a category $\mathcal{C}$ and an object $U \in \text{obj}(\mathcal{C})$, we will simply write U instead of $(\mathcal{C}/U)$ to denote the slice category over U .
- We know from 2.1 that a (pre)sheaf $X : \mathcal{C}^{op} \rightarrow (\text{Sets})$ can induce a CFG $X \rightarrow \mathcal{C}$ which we can also denote by X .

Definition 3.9 (Algebraic Space). Let S be a scheme. An **algebraic space** over S is a sheaf X on $(\text{Sch}/S)_{\text{ét}}$ such that the following two conditions are satisfied:

1. The diagonal morphism over (Sch) $\Delta : X \rightarrow X \times X$ is representable.
2. There exists a scheme Y and a surjective étale morphism $Y \rightarrow X$.

Remark. Recall the remark after Definition 3.4 that we can introduce various notions of morphisms via the Cartesian diagram. The statement that $Y \rightarrow X$ is a surjective étale

morphism means that for all schemes U and all morphisms $U \rightarrow X$, the projection morphism h in the following Cartesian diagram is a surjective étale morphism of schemes:

$$\begin{array}{ccc} Y \times_X U & \longrightarrow & Y \\ h \downarrow & & \downarrow \\ U & \longrightarrow & X \end{array}$$

Note that the fiber product $Y \times_X U$ is indeed guaranteed to be a scheme because we assumed in condition (1) that the diagonal Δ is representable by schemes. There is a fact that we can consider $Y \times_X U$ as $X \times_{X \times X} (Y \times U)$.

Example 3.10. Consider a space defined as a quotient $X \cong Y/R$ by an étale equivalence relation. This space naturally admits an étale cover $Y \rightarrow X$, which means condition (2) of the definition is satisfied.

To verify condition (1), one must analyze the diagonal $\Delta : X \rightarrow X \times X$ via the following Cartesian diagram:

$$\begin{array}{ccc} R & \longrightarrow & Y \times Y \\ \downarrow & & \downarrow \\ X & \xrightarrow{\Delta} & X \times X \end{array}$$

The detailed verification that this implies Δ is representable by schemes is omitted here.

Example(An algebraic space that is not a scheme): A natural question arises: does there exist an algebraic space that is not a scheme?

Yes, there exists such an example in dimension no less than 3, originally constructed by Hironaka.

4 2026.3.23

4.1 Examples

Example 4.1. Let X be a scheme and G be a finite group acting on X . We can define a presheaf quotient $(X/G)^{\text{pre}} : \text{Sch}^{\text{op}} \rightarrow \text{Sets}$ by $T \mapsto X(T)/G$. Let X/G be its sheafification in $\text{Sh}_{\text{et}}(\text{Sch})$.

Question: Is X/G a scheme?

First, let us consider what properties X/G must satisfy if it is indeed a scheme.

Suppose X/G is a scheme. Then we have a projection morphism $\pi : X \rightarrow X/G$. (Note that if $X = \text{Spec} A$, this is just the finite morphism $\text{Spec} A \rightarrow \text{Spec} A^G$). For any point $[x] \in X/G$, there exists an affine open neighborhood U containing $[x]$. Since π is a finite morphism, the preimage $\pi^{-1}(U)$ is also affine, and it clearly contains the orbit $G \cdot x$.

Theorem 4.2 (From SGA). The quotient X/G is a scheme if and only if for every $x \in X$, the orbit $G \cdot x$ is contained in an open affine subscheme of X .

As an application, suppose $X \subseteq \mathbb{P}_k^n$ is a projective scheme over an infinite field k . Since the orbit $G \cdot x$ is a finite set, we can always find a hyperplane $H \cong \mathbb{P}^{n-1}$ such that $H \cap (G \cdot x) = \emptyset$. Then $X \setminus (H \cap X)$ is a closed subscheme of $\mathbb{P}^n \setminus H \cong \mathbb{A}^n$, which means $X \setminus (H \cap X)$ is affine. Therefore, the orbit $G \cdot x$ is contained in $X \setminus (H \cap X)$, which is an open affine subscheme of X . By the theorem above, this implies that X/G is a scheme.

Example 4.3. Let us detail the construction of Hironaka's example^a of a non-projective smooth 3-fold and a non-scheme quotient. Assume $\dim X = 3$ and X contains two curves C and D intersecting at two points p and q .

^aYou can refer to GTM52 page 443.

Around the point p , first we blow up C , and then blow up D (strictly speaking, we blow up the strict transform of D). Symmetrically, around the point q , we first blow up D , and then blow up the strict transform of C . We glue these local constructions over $X \setminus \{p, q\}$ to obtain a new smooth scheme $\tilde{X}$. The involution $\iota : X \rightarrow X$ lifts to an involution $\tilde{\iota} : \tilde{X} \rightarrow \tilde{X}$.

Let $\pi : \tilde{X} \rightarrow X$ be the blow-down morphism. Let H_C and H_D be the exceptional divisors (which have dimension 2) over the curves C and D (which have dimension 1).

Let us analyze the fibers over the intersection points p and q . Over p , the fiber $\pi^{-1}(p)$

consists of two curves, say l_p and $l_{p'}$. We have the following numerical equivalences:

$$\begin{aligned}\pi^{-1}(p) &= l_p + l_{p'} \equiv \pi^{-1}(x) \\ l_{p'} &\equiv \pi^{-1}(y)\end{aligned}$$

where x is a general point on C and y is a general point on D . Similarly, over q , the fiber $\pi^{-1}(q)$ consists of two curves l_q and $l_{q'}$, satisfying:

$$\begin{aligned}\pi^{-1}(q) &= l_q + l_{q'} \equiv \pi^{-1}(y) \\ l_{q'} &\equiv \pi^{-1}(x)\end{aligned}$$

From these relations, we can deduce that:

$$l_p + l_q \equiv 0 \quad (\text{as a 1-cycle})$$

This immediately shows that $\tilde{X}$ is not projective. If $\tilde{X}$ were projective, there would exist an ample divisor H . Intersecting H with our curves would give $H \cdot l_p > 0$ and $H \cdot l_q > 0$, which implies $H \cdot (l_p + l_q) > 0$. This is a contradiction since $l_p + l_q \equiv 0$.

Goal: Prove that the quotient space $\tilde{X}/\tilde{i}$ is not a scheme.

Suppose, for the sake of contradiction, that $\tilde{X}/\tilde{i}$ is a scheme. Take a point $\alpha \in l_p$. Its image under the involution is $\tilde{i}(\alpha) \in l_q$.

By the property of scheme quotients as discussed previously, there must exist an affine open neighborhood U in $\tilde{X}$ that contains the orbit $\{\alpha, \tilde{i}(\alpha)\}$. Thus, both $\alpha \in U$ and $\tilde{i}(\alpha) \in U$.

Note that l_p and l_q are projective curves arising from blow-ups, so they cannot be entirely contained in an affine scheme¹. Therefore, $l_p \not\subset U$ and $l_q \not\subset U$. This means both curves must intersect the closed complement $U^c = \tilde{X} \setminus U$.

Since $\tilde{X}$ is a smooth variety, the complement of an affine open subset has pure codimension 1². Thus, the irreducible components of $\tilde{X} \setminus U$ form the support of an effective Cartier divisor $D = \tilde{X} \setminus U$.

Because l_p and l_q intersect D but are not contained in it, their intersection numbers with D must be strictly positive:

$$D \cdot l_p > 0 \quad \text{and} \quad D \cdot l_q > 0$$

¹Otherwise, from the fact that it is a closed subscheme of an affine scheme, we would deduce that l_p (l_q) is affine, which is a contradiction since they are projective curves by analyzing $\Gamma(l_p, \mathcal{O}_{l_p})$.

²Sketch of proof: Otherwise we can find an affine open subscheme $V \subseteq \tilde{X}$ such that $V \cap (\tilde{X} \setminus U) \neq \emptyset$ is irreducible, $\text{codim}_V V \cap (\tilde{X} \setminus U) \geq 2$, $U \cap V \neq \emptyset$. By separatedness $U \cap V$ is again affine. By Hartog's Theorem (reference: Gortz's book I Theorem 6.45) $V = U \cap V$ which contradicts to the choice of V .

Consequently, $D \cdot (l_p + l_q) > 0$. However, this contradicts the numerical equivalence $l_p + l_q \equiv 0$ established earlier.

Therefore, no such affine open set U can exist, which implies that $\tilde{X}/\tilde{I}$ is not a scheme.

4.2 Simplification of the definition of Algebraic Spaces

Definition 4.4 (Algebraic Space). An **algebraic space** is a sheaf X on $(\text{Sch})_{\text{ét}}$ such that there exists a scheme Y and a surjective étale morphism $Y \rightarrow X$.

Recall that in the original definition of algebraic spaces, we required diagonal morphism $X \rightarrow X \times X$ to be representable. We will show that this condition is automatically satisfied for any sheaf X on $(\text{Sch})_{\text{ét}}$ that admits a surjective étale morphism from a scheme.

Theorem 4.5. The diagonal morphism $X \rightarrow X \times X$ of an algebraic space is representable.

Proof Sketch. By the definition of an algebraic space, there exists an étale atlas $U \rightarrow X$ where U is a scheme. Let $R = U \times_X U$, which is also a scheme by definition.

Recall that the morphism $X \rightarrow X \times X$ is representable if and only if for any scheme T and any morphism $T \rightarrow X \times X$, the fiber product $Y = T \times_{X \times X} X$ is a scheme.

To show this, we construct a base change diagram. Let $T' = T \times_{X \times X} (U \times U)$. Since $U \rightarrow X$ is an étale atlas, the induced morphism $U \times U \rightarrow X \times X$ is an étale cover. Consequently, the base change $T' \rightarrow T$ is also an étale cover, which implies that T' is a scheme.

Next, we define $Y' = Y \times_T T'$. By the universal property of fiber products, we can fit these spaces into the following commutative cube, where all faces are Cartesian squares:

$$\begin{array}{ccccc}
 Y' & \xrightarrow{\quad} & T' & & \\
 \downarrow & \searrow & \downarrow & \searrow & \\
 & R & \xrightarrow{\quad} & U \times U & \\
 \downarrow & \downarrow & \downarrow & \downarrow & \\
 Y & \xrightarrow{\quad} & T & & \\
 \downarrow & \searrow & \downarrow & \searrow & \\
 & X & \xrightarrow{\quad} & X \times X &
 \end{array}$$

From the top face of the cube, we can see that Y' is the pullback of R along the morphism $T' \rightarrow U \times U$. Since R , $U \times U$, and T' are all schemes, it follows that Y' is a scheme. We also

have the Cartesian square for the étale cover:

$$\begin{array}{ccc} Y' & \longrightarrow & T' \\ \downarrow & & \downarrow \\ Y & \longrightarrow & T \end{array}$$

Now, observe that the diagonal morphism $X \rightarrow X \times X$ is always a monomorphism. Since $R \rightarrow U \times U$ is the base change of the diagonal $X \rightarrow X \times X$ along the cover $U \times U \rightarrow X \times X$, it follows that the morphism $R \rightarrow U \times U$ is also a monomorphism.

By a standard result in scheme theory (see Stacks Project, Tag 01TC), a monomorphism between schemes is locally quasi-finite and separated. Therefore, the morphism $R \rightarrow U \times U$ is locally quasi-finite and separated.

Since the property of being locally quasi-finite and separated is stable under base change, and $Y' \rightarrow T'$ is the base change of $R \rightarrow U \times U$, we conclude that the morphism $Y' \rightarrow T'$ is also locally quasi-finite and separated.

We can finish the proof by applying the following proposition: □

Proposition 4.6. Let $Y \in \mathrm{Sh}_{\text{ét}}(\mathrm{Sch})$. Suppose we have a Cartesian square:

$$\begin{array}{ccc} Y' & \longrightarrow & T' \\ \downarrow & & \downarrow_{\text{fppf}} \\ Y & \longrightarrow & T \end{array}$$

where the morphism $T' \rightarrow T$ is fppf (faithfully flat and of finite presentation), $Y' \rightarrow T'$ is locally quasi-finite and separated. If Y' is a scheme, then Y is also a scheme.

Theorem 4.7 (Affine Descent Theorem). Let $Y \in \mathrm{Sh}_{\text{ét}}(\mathrm{Sch})$. Suppose we have a Cartesian square:

$$\begin{array}{ccc} Y' & \longrightarrow & T' \\ \downarrow & & \downarrow_{\text{fppf}} \\ Y & \longrightarrow & T \end{array}$$

where the morphism $T' \rightarrow T$ is fppf (faithfully flat and of finite presentation), $Y' \rightarrow T'$ is affine. If Y' is a scheme and satisfies the following conditions:

- We have an isomorphism α between the pullbacks:

$$\begin{array}{ccccc}
 & & p_1^* Y' & \longrightarrow & Y' \\
 & \swarrow \exists \alpha, \simeq & \downarrow & & \downarrow \\
 p_2^* Y' & \longrightarrow & T' \times_T T' & \xrightarrow{p_1} & T' \\
 \downarrow & & \downarrow p_2 & & \downarrow \\
 Y' & \longrightarrow & T' & \longrightarrow & T
 \end{array}$$

- Furthermore, the isomorphism α satisfies the cocycle condition.

Then Y is a scheme.

4.3 Comparing with Schemes

Theorem 4.8. Let X be a quasi-compact and quasi-separated algebraic space. Then there exists an open dense subspace $U \subseteq X$ such that U is a scheme.

Proof Sketch. Let $f : V \rightarrow X$ be an étale atlas where V is an affine scheme. Since X is quasi-compact and quasi-separated, the morphism f is quasi-compact. This implies that there exists an open dense subspace $U \subseteq X$ such that the restriction $f|_{f^{-1}(U)} : f^{-1}(U) \rightarrow U$ is a finite morphism.

By passing to a Galois cover, we can assume that U is isomorphic to a quotient W/G , where W is a scheme and G is a finite group.

We can find a G -invariant quasi-affine open subset $W_1 \subseteq W$ by taking an arbitrary quasi-affine open subset and intersecting it with all its translations under the group action, i.e., $W_1 = \bigcap_{g \in G} g \cdot W_0$. Similarly, within W_1 , we can find a G -invariant affine open subset $W_2 \subseteq W_1$ by taking the intersection of the G -translates of an affine open subset.

Since W_2 is an affine scheme and G is a finite group acting on it, the quotient space W_2/G is a scheme. Therefore, W_2/G forms an open dense subscheme of U , which completes the proof. $\square$

Theorem 4.9. Let X be a separated, Noetherian algebraic space of finite type.

1. If the dimension of X is less than or equal to 1, then X is a scheme.
2. If X is a regular, integral algebraic space of finite type over a field k with dimension 2, then X is a scheme.

Theorem 4.10 (Artin's Contraction Theorem). Let X be an algebraic surface and $C \subseteq X$ be a curve such that $C \cdot C < 0$. Then there exists a contraction morphism $X \rightarrow T$ that contracts the curve C to a point $p \in T$.

Remark. *In the complex analytic category, a theorem by Grauert states that there exists an analytic space T that achieves this contraction. In the algebraic category, Artin proved that there exists an algebraic space T that achieves this contraction.*

5 2026.3.25

5.1 Review

Let X be an algebraic space, which can be viewed as a functor $X : \text{Sch}^{\text{op}} \rightarrow \text{Sets}$. Consider a morphism $\text{Spec}K \xrightarrow{p} X$, which corresponds to an element in $X(\text{Spec}K)$.

We define an equivalence relation as follows: $(\text{Spec}K \xrightarrow{p} X) \sim (\text{Spec}K' \xrightarrow{p'} X)$ if and only if there exists a field Ω and field extensions $K \hookrightarrow \Omega$ and $K' \hookrightarrow \Omega$ such that the following diagram commutes:

$$\begin{array}{ccc} \text{Spec}\Omega & \longrightarrow & \text{Spec}K \\ \downarrow & & \downarrow p \\ \text{Spec}K' & \xrightarrow{p'} & X \end{array}$$

Definition 5.1. The **underlying topological space** $|X|$ of an algebraic space X is defined as the set of equivalence classes:

$$|X| = \{\text{Spec}K \rightarrow X\} / \sim.$$

The topology on $|X|$ is defined by the condition that a subfunctor $U \hookrightarrow X$ is an open algebraic subspace if and only if $|U| \subseteq |X|$ is an open subset.

Lemma 5.2. Let $U \twoheadrightarrow X$ be an étale surjective morphism. Then the induced map $|U| \rightarrow |X|$ is a continuous, open, and surjective map with respect to the Zariski topology.

Lemma 5.3. If X is a scheme, then $|X|$ coincides with its usual underlying topological space.

Remark. Recall from the previous lecture regarding the term “dense”: being dense is not a representable property. Rather, it is an additionally defined concept that refers specifically to a property of the underlying topological space $|X|$.

5.2 Deligne-Mumford Stacks

Motivation: Many moduli functors $M : \text{Sch}^{\text{op}} \rightarrow \text{Sets}$ are not representable.

Slogan: A Deligne-Mumford (DM) stack is a stack such that its objects have finite automorphism groups, i.e., $|\text{Aut}| < +\infty$.

To work with stacks, we enlarge our category by considering the natural inclusion of functor

categories:

$$\mathrm{Fun}(\mathrm{Sch}^{\mathrm{op}}, \mathrm{Sets}) \hookrightarrow \mathrm{Fun}(\mathrm{Sch}^{\mathrm{op}}, \mathrm{Groupoids}).$$

Definition 5.4. Let $\mathcal{X}$ and $\mathcal{Y}$ be prestacks over the category of schemes (Sch) , equipped with structure functors $p : \mathcal{X} \rightarrow (\mathrm{Sch})$ and $q : \mathcal{Y} \rightarrow (\mathrm{Sch})$.

1. A functor $f : \mathcal{X} \rightarrow \mathcal{Y}$ is called a **1-morphism** if it strictly commutes with the projections to the base category (Sch) .

$$\begin{array}{ccc} \mathcal{X} & \xrightarrow{f} & \mathcal{Y} \\ & \searrow p & \swarrow q \\ & (\mathrm{Sch}) & \end{array}$$

Explicitly, for all objects $a, b \in \mathcal{X}$ and any morphism $a \rightarrow b$ in $\mathcal{X}$, we require that $p(a) = q(f(a))$, $p(b) = q(f(b))$, and that the induced morphisms over the base scheme satisfy $p(a \rightarrow b) = q(f(a) \rightarrow f(b))$.

Alternatively, viewing prestacks as pseudo-functors $\mathcal{X}, \mathcal{Y} : \mathrm{Sch}^{\mathrm{op}} \rightarrow \mathrm{Groupoids}$, a 1-morphism consists of a functor $f_S : \mathcal{X}(S) \rightarrow \mathcal{Y}(S)$ for each scheme S .

2. Let $f, g : \mathcal{X} \rightarrow \mathcal{Y}$ be two 1-morphisms. A **2-morphism** $\alpha : f \Rightarrow g$ is a natural transformation such that for all objects $a \in \mathcal{X}$, the morphism $\alpha_a : f(a) \rightarrow g(a)$ in $\mathcal{Y}$ lies over the identity morphism on the base scheme $S = p(a)$.

From the pseudo-functor perspective, for all objects $a \in \mathcal{X}(S)$, given the functors $f_S, g_S : \mathcal{X}(S) \rightarrow \mathcal{Y}(S)$, the component $\alpha_{a,S} : f_S(a) \xrightarrow{\sim} g_S(a)$ is necessarily an isomorphism because the target category $\mathcal{Y}(S)$ is a groupoid.

3. A 1-morphism $f : \mathcal{X} \rightarrow \mathcal{Y}$ is an **isomorphism** if there exists a 1-morphism $g : \mathcal{Y} \rightarrow \mathcal{X}$ along with 2-morphisms $f \circ g \Rightarrow \mathrm{id}_{\mathcal{Y}}$ and $g \circ f \Rightarrow \mathrm{id}_{\mathcal{X}}$.

Definition 5.5. A 1-morphism $f : X \rightarrow Y$ between stacks is said to be **representable** if, for all schemes T and any morphism $T \rightarrow Y$, the fiber product $X \times_Y T$ is an algebraic space.

$$\begin{array}{ccc} X \times_Y T & \longrightarrow & T \\ \downarrow & & \downarrow \\ X & \xrightarrow{f} & Y \end{array}$$

Furthermore, we say that a representable morphism $X \rightarrow Y$ has a property P if, for any

scheme $T \rightarrow Y$, the composition $U \rightarrow X \times_Y T \rightarrow T$ has property P , where $U \rightarrow X \times_Y T$ is an étale atlas. Note that the morphism $X \times_Y T \rightarrow T$ is not represented by schemes in general.

Definition 5.6. A **Deligne-Mumford (DM) stack** (resp. **algebraic stack**) is a stack X over the category of schemes such that there exists a scheme U and an étale (resp. smooth) surjective morphism $U \rightarrow X$.

Remark. We have the following observations regarding the definition:

1. The definition implies that the diagonal morphism $X \rightarrow X \times X$ is representable.
2. An algebraic stack X is a DM stack if and only if the diagonal morphism $X \rightarrow X \times X$ is unramified. This is equivalent to the condition that for any object x , its automorphism group $\text{Aut}(x)$ —which can be viewed as the pullback of the diagonal $\Delta^*(x, x)$ —is discrete and reduced.

5.3 Discussion about Quotient Stacks

Let X be a scheme and G be a smooth affine algebraic group acting on X . We first define the prestack quotient

$$[X/G]^{\text{pre}} : \text{Sch}^{\text{op}} \rightarrow \text{Groupoids}$$

by assigning to each scheme T the quotient groupoid $[X(T)/G(T)]$.

Then, by taking the stackification, we obtain the quotient stack $[X/G] : \text{Sch}^{\text{op}} \rightarrow \text{Groupoids}$. For a scheme T , the objects of the groupoid $[X/G](T)$ are given by diagrams of the following form:

$$\begin{array}{ccc} P & \xrightarrow{G\text{-equiv}} & X \\ G\text{-torsor} \downarrow & & \\ T & & \end{array}$$

where $P \rightarrow T$ is a principal G -bundle (or G -torsor) over T , and $P \rightarrow X$ is a G -equivariant morphism.

Theorem 5.7. Let X be an algebraic space over a base scheme S , and let G be a smooth affine group scheme over S acting on X . Then the quotient stack $[X/G]$ is an algebraic stack. Furthermore, if G is an étale group scheme over S , then $[X/G]$ is a DM stack.

Remark. When G is a finite group, the associated group scheme over S is $G_S = \coprod_{g \in G} S$. In this case, the quotient stack $[X/G]$ mentioned in previous contexts is exactly the stack $[X/G_S]$ defined here.

Proof Sketch. Recall that a morphism from a scheme T to the quotient stack $[X/G]$ corresponds to a groupoid whose objects are diagrams of the form:

$$\left\{ \begin{array}{ccc} P & \longrightarrow & X \\ \downarrow & & \\ T & & \end{array} \right\}$$

where $P \rightarrow T$ is a principal G -bundle (a G -torsor) and $P \rightarrow X$ is a G -equivariant morphism.

Consider the trivial G -torsor $G \times X \rightarrow X$ over X . The group action provides a natural G -equivariant morphism $G \times X \rightarrow X$. This data induces a natural morphism $X \rightarrow [X/G]$.

We have the following conclusion. It implies that $X \rightarrow [X/G]$ is representable, and étaleness (resp. smoothness) of $X \rightarrow [X/G]$ follows from étaleness (resp. smoothness) of every $P \rightarrow T$. $\square$

Exercise 5.8. Given any morphism $T \rightarrow [X/G]$ from a scheme T , show that we have the following Cartesian square (fiber product):

$$\begin{array}{ccc} P & \longrightarrow & X \\ \downarrow & & \downarrow \pi \\ T & \longrightarrow & [X/G] \end{array}$$

where $\pi : X \rightarrow [X/G]$ is the natural projection defined above.

Corollary 5.9. Suppose G is a finite group and the action of G on X is free. Then the quotient stack $[X/G]$ is an algebraic space.

Proof. By definition, the action of G on X is free if the map $G \times X \rightarrow X \times X$, given by $(g, x) \mapsto (gx, x)$, is a monomorphism.

To show that the stack $[X/G]$ is an algebraic space, we want to show that for any scheme T , the groupoid $[X/G]^{\text{pre}}(T)$ is equivalent to a set. This means that for all objects $x, y \in X(T)$ equipped with the $G(T)$ -action, there exists either exactly 0 or exactly 1 morphism from x to y .

Suppose for the sake of contradiction that there are 2 morphisms from x to y . These correspond to elements $g, h \in G(T)$ such that $gx = y$ and $hx = y$. This implies that under the

map $G \times X \rightarrow X \times X$, we have:

$$\begin{aligned}(g, x) &\mapsto (y, x) \\ (h, x) &\mapsto (y, x)\end{aligned}$$

Because the map $G \times X \rightarrow X \times X$ is a monomorphism, it follows that $(g, x) = (h, x)$, and therefore $g = h$.

Thus, there is at most one morphism between any two objects in the prestack category. Consequently, the stackification $[X/G]$ is equivalent to a sheaf of sets, meaning it is an algebraic space. $\square$

5.4 *Example: The Moduli of Elliptic Curves

Consider elliptic curves defined by the Weierstrass equation $E = \{y^2 = x^3 + ax + b\} \subset \mathbb{A}^2$. By homogenizing, we obtain a projective curve $\bar{E} \subseteq \mathbb{P}^2$, with a distinguished base point $[0 : 1 : 0] = O$ on E .

The curve is nonsingular if and only if the discriminant $\Delta = 4a^3 + 27b^2 \neq 0$. Thus, the parameter space for such curves is the open subscheme $(\mathbb{A}_{a,b}^2 \setminus \Delta)$ over $\text{Spec}(\mathbb{C})$.

There is an action of the multiplicative group $\mathbb{G}_m$ on this parameter space. For $\lambda \in \mathbb{G}_m$, the transformation $x \mapsto \lambda^2 x$ and $y \mapsto \lambda^3 y$ induces an isomorphism between the curve with parameters (a, b) and the curve with parameters (a', b') , where:

$$a' = \lambda^4 a, \quad b' = \lambda^6 b.$$

This gives an isomorphism $E_{a,b} \simeq E_{a',b'}$. Therefore, we have a group action $\mathbb{G}_m \curvearrowright (\mathbb{A}_{a,b}^2 \setminus \Delta)$.

Taking the quotient, we obtain a bijection at the level of complex points:

$$(\mathbb{A}_{a,b}^2 \setminus \Delta) / \mathbb{G}_m(\text{Spec}(\mathbb{C})) \xrightarrow{\sim} \mathbb{A}^1(\mathbb{C})$$

given by mapping the class of (a, b) to the j -invariant:

$$j = 1728 \cdot \frac{4a^3}{\Delta}.$$

Remark (Analytic Perspective). *Over the complex numbers, an elliptic curve can be viewed as a torus $E \simeq \mathbb{C}/\Lambda$, where the lattice is $\Lambda = \mathbb{Z} \cdot 1 + \mathbb{Z} \cdot \tau$ for some τ in the upper half-plane $\mathbb{H}$. We have a sequence of maps:*

$$\mathbb{H} \longrightarrow \mathbb{H}/\text{SL}_2(\mathbb{Z}) \longrightarrow \mathbb{C}.$$

There is a universal family of elliptic curves $\mathcal{E}$ over $\mathbb{H}$. However, this universal family cannot

descend to the quotient space $\mathbb{H}/\mathrm{SL}_2(\mathbb{Z})$. The reason is the existence of extra automorphisms at the points corresponding to $j = 0$ and $j = 1728$.

Non-representability by Schemes

Let us define the moduli functor of elliptic curves $\mathcal{M} : \mathrm{Sch}^{\mathrm{op}} \rightarrow \mathrm{Sets}$ by assigning to each scheme S the set of isomorphism classes of families of elliptic curves over S :

$$\mathcal{M}(S) = \left\{ \mathcal{E} \xrightarrow{\pi} S \text{ family of elliptic curves} \right\} / \sim.$$

Proposition 5.10. The functor $\mathcal{M}$ is not representable by a scheme.

Proof. We will prove this by constructing a family $\mathcal{E} \xrightarrow{\pi} S$ such that all its fibers $\mathcal{E}_s$ are isomorphic to a fixed elliptic curve E , but the family $\mathcal{E} \rightarrow S$ itself is non-trivial.

Let the base space be $S = \mathbb{A}^1 \setminus \{0\}$ with coordinate t . We construct a family $\mathcal{E} \subset (\mathbb{A}^1 \setminus \{0\}) \times \mathbb{P}^2$ given by the equation:

$$\mathcal{E} = V(y^2z - x^3 - tz^3).$$

This corresponds to setting $a = 0$ and $b = t$. For any $t \in \mathbb{A}^1 \setminus \{0\}$, the fiber E_t has j -invariant $j = 0$. Therefore, for any $t_1, t_2 \in S$, the fibers are isomorphic: $E_{t_1} \simeq E_{t_2}$.

Now, assume for the sake of contradiction that $\mathcal{M}$ is representable by a scheme M . Then, by Yoneda's lemma, $\mathcal{M}(S) \simeq \mathrm{Hom}(S, M)$. The family $\mathcal{E} \rightarrow S$ corresponds to a classifying morphism $S \rightarrow M$.

Since all fibers of $\mathcal{E}$ over S are isomorphic to the same curve (with $j = 0$), the classifying morphism $S \rightarrow M$ must map all of S to a single point in M (the point corresponding to $j = 0$). This means the morphism factors through a point $\mathrm{Spec}(\mathbb{C})$:

$$\begin{array}{ccc} S & \xrightarrow{\quad} & M \\ \downarrow & \nearrow & \\ \mathrm{Spec}(\mathbb{C}) & & \end{array}$$

Because the map factors through a point, the family $\mathcal{E} \rightarrow S$ must be the pullback of a family over a point. This implies that $\mathcal{E} \rightarrow S$ must be a trivial family (i.e., isomorphic to $S \times E_1$). However, one can show that our constructed family $\mathcal{E} \rightarrow S$ is a non-trivial twist, which yields a contradiction. Thus, $\mathcal{M}$ cannot be a scheme. $\square$

The Stack Perspective

Because the functor is not representable, we must view $\mathcal{M}$ as a stack. We can explicitly describe this stack as the quotient stack $[(\mathbb{A}^2 \setminus \Delta)/\mathbb{G}_m]$.

Given a family of elliptic curves $\mathcal{E} \xrightarrow{\pi} T$, we can consider the pushforward of the relative cotangent bundle, $\pi_*\Omega_{\mathcal{E}/T}^1$. Because the fibers are elliptic curves, $H^0(E_t, \Omega^1) = 1$, so this pushforward is a line bundle L on T .

By removing the zero section from L , we obtain $L \setminus \{\text{zero}\} \xrightarrow{p} T$, which is a principal $\mathbb{G}_m$ -bundle (a $\mathbb{G}_m$ -torsor) over T . Let us denote this torsor by P . The original line bundle can be recovered as the associated bundle $L = P \times_{\mathbb{G}_m} \mathbb{A}^1$, where the equivalence relation is given by $(p \cdot \lambda, a) \sim (p, \lambda a)$.

This $\mathbb{G}_m$ -torsor provides the necessary rigidification to map the family to the parameter space $\mathbb{A}^2 \setminus \Delta$, demonstrating that the moduli stack is equivalent to the quotient stack:

$$\mathcal{M} \simeq [(\mathbb{A}^2 \setminus \Delta)/\mathbb{G}_m].$$

Finally, we note that this is a Deligne-Mumford stack because the automorphism groups of the objects are finite. Specifically, for an elliptic curve E_j with a given j -invariant, the automorphism group is:

$$\text{Aut}(E_j) = \begin{cases} \mathbb{Z}/2\mathbb{Z} & \text{if } j \neq 0, 1728, \\ \mathbb{Z}/4\mathbb{Z} & \text{if } j = 1728, \\ \mathbb{Z}/6\mathbb{Z} & \text{if } j = 0. \end{cases}$$

Since $|\text{Aut}(x)| < +\infty$ for all objects, the stack is Deligne-Mumford.

5.5 Coarse Moduli Space

Definition 5.11. A morphism $\pi : \mathcal{X} \rightarrow X$ from an algebraic stack $\mathcal{X}$ to an algebraic space X is called a **coarse moduli space** if it satisfies the following two conditions:

1. For all algebraically closed fields $K = \bar{K}$, the induced map on isomorphism classes of objects

$$\mathcal{X}(\text{Spec}K)/\sim \longrightarrow X(\text{Spec}K)$$

is a bijection.

2. The morphism π is universal for maps to algebraic spaces. That is, for any algebraic space Y and any morphism $\mathcal{X} \rightarrow Y$, there exists a unique morphism $X \rightarrow Y$ such

that the following diagram commutes:

$$\begin{array}{ccc} \mathcal{X} & \longrightarrow & Y \\ \pi \downarrow & \nearrow \exists! & \\ X & & \end{array}$$

Theorem 5.12 (Keel-Mori). Let S be a Noetherian scheme. Let $\mathcal{X}$ be a Deligne-Mumford stack over S that is separated and of finite type. Then there exists a coarse moduli space $\pi : \mathcal{X} \rightarrow X$ satisfying the following properties:

1. $\mathcal{O}_X = \pi_* \mathcal{O}_{\mathcal{X}}$.
2. The algebraic space X is separated and of finite type over S .
3. The morphism π is proper and a universal homeomorphism.
4. The stack $\mathcal{X}$ is proper over S if and only if the algebraic space X is proper over S .

6 2026.3.30

Example 6.1. Here are some fundamental examples of algebraic stacks:

1. Algebraic spaces are DM stacks.
2. Consider the quotient stack $[U/G]$, where G is an algebraic group acting algebraically on a scheme $U \in \text{Sch}$.
 - If G is a finite group, then the quotient stack $[U/G]$ is a DM stack.
 - If G is a smooth group, then the quotient stack $[U/G]$ is an Artin stack.

We know that a scheme is constructed by gluing affine schemes together Zariski locally. A natural question arises: how do we view an algebraic stack locally?

In fact, from a local perspective, we have the following descriptions:

- An **Artin stack** $\mathcal{X}$ is locally (in the smooth topology) isomorphic to a quotient stack $[U/G]$, where G is a smooth algebraic group.
- A **DM stack** is locally (in the étale topology) isomorphic to a quotient stack $[U/G]$, where G is a finite reduced group scheme.

6.1 Stabilizer and Inertia Stacks

Definition 6.2 (Stabilizer). Let $\mathcal{X}$ be an algebraic stack over the category of schemes Sch_S . Given a point $x : \text{Spec}(k) \rightarrow \mathcal{X}$, the **stabilizer** G_x is defined as the automorphism group of x , namely $G_x \triangleq \text{Aut}_{\mathcal{X}(k)}(x)$.

Equivalently, this can be expressed as the fiber product:

$$G_x \cong \text{Spec}(k) \times_{\mathcal{X} \times \mathcal{X}} \mathcal{X}$$

Here, the morphism $\text{Spec}(k) \rightarrow \mathcal{X} \times \mathcal{X}$ is given by the pair (x, x) , and the morphism $\mathcal{X} \rightarrow \mathcal{X} \times \mathcal{X}$ is the diagonal morphism Δ .

Definition 6.3 (Inertia Stack). The **inertia stack** of $\mathcal{X}$, denoted by $I_{\mathcal{X}}$, is defined as the fiber product:

$$I_{\mathcal{X}} \triangleq \mathcal{X} \times_{\mathcal{X} \times \mathcal{X}} \mathcal{X}$$

For any scheme U , the objects of the groupoid $I_{\mathcal{X}}(U)$ consist of pairs (x, α) , where $x \in \mathcal{X}(U)$ is an object and $\alpha \in \text{Aut}_{\mathcal{X}(U)}(x)$ is an automorphism of x .

Remark. *There exists a natural forgetful morphism $\pi : I_{\mathcal{X}} \rightarrow \mathcal{X}$, which maps a pair (x, α) to x . (Strictly speaking, this is a slight abuse of notation, but it captures the geometric intuition). The fiber of this morphism π over a point x is precisely the stabilizer G_x .*

A natural question arises: when is the morphism π an equivalence? We have the following facts regarding the properties of π and the structure of the stack $\mathcal{X}$.

Proposition 6.4. Let $\pi : I_{\mathcal{X}} \rightarrow \mathcal{X}$ be the forgetful morphism from the inertia stack.

1. If π is a monomorphism, then $\mathcal{X}$ is an algebraic space.
2. $\mathcal{X}$ is a DM stack if and only if the fibers of π are finite and reduced group algebraic spaces.

Example 6.5. Here are some concrete examples of inertia stacks^a:

1. Let $\mathcal{X} = BG'$ be the classifying stack of an algebraic group G' (i.e., the category of G' -bundles), which is by definition $[\text{pt}/G']$. For any scheme U , the objects $\mathcal{X}(U)$ are G' -torsors $P \rightarrow U$. The inertia stack is given by the quotient stack:

$$I_{\mathcal{X}} \cong [G'/G']$$

where G' acts on itself by conjugation. For any point $x : \text{Spec}(k) \rightarrow BG'$, the stabilizer is $G_x \cong G'(k)$.

2. Let $\mathcal{X} = [U/G']$ be a quotient stack. The inertia stack is given by:

$$I_{\mathcal{X}} = \mathcal{X} \times_{\mathcal{X} \times \mathcal{X}} \mathcal{X} \cong [Z/G']$$

where $Z = \{(u, g) \in U \times G' : g \cdot u = u\}$. Given a point $u \in U$, which corresponds to a composition of morphisms $\text{Spec}(k) \xrightarrow{u} U \rightarrow [U/G']$, the stabilizer G_u is exactly the isotropy group $G'_u = \{g \in G' : g \cdot u = u\}$ of u under the action of G' .

3. Let $\mathcal{X} = \mathcal{M}_{1,1}$ be the moduli stack of elliptic curves. For any point $x : \text{Spec}(k) \rightarrow \mathcal{X}$, it corresponds to an elliptic curve (E, O) over the field k , where $O \in E$ is the origin.

The stabilizer $G_x = \text{Aut}_k(E, O)$ is given by the following cases:

$$G_x = \begin{cases} \{\pm 1\} & \text{for a general } (E, O), \\ \mathbb{Z}/4\mathbb{Z} & \text{if the } j\text{-invariant } j(E, O) = 1728 \text{ (e.g., } y^2 = x^3 + x), \\ \mathbb{Z}/6\mathbb{Z} & \text{if the } j\text{-invariant } j(E, O) = 0 \text{ (e.g., } y^2 = x^3 + 1). \end{cases}$$

So by the previous Proposition, $\mathcal{M}_{1,1}$ is a DM stack.

^aYou can see the inertia stack by the second view of it in the definition of the inertia stack.

Remark. We have the following observations regarding quotient stacks and DM stacks:

1. A quotient stack $[U/G]$, where the algebraic group G is not necessarily finite, is a DM stack if the stabilizer G_u is finite and reduced for all points $u \in U$.
2. Consider the classifying stack $B\mu_p$ over the category of schemes over $\mathbb{F}_p$, denoted as $\text{Sch}_{\mathbb{F}_p}$. For any scheme $U \in \text{Sch}_{\mathbb{F}_p}$, the objects of $B\mu_p(U)$ are μ_p -torsors $P \rightarrow U$. For any point $x : \text{Spec}(k) \rightarrow B\mu_p$, the stabilizer G_x is a finite group, but it is not reduced. Recall that $\mu_p = \text{Spec}(\mathbb{F}_p[t]/(t^p - 1))$. (Note: Here, the base field $\mathbb{F}_p$ can be replaced by any field of characteristic p). Because the stabilizers are not reduced, $B\mu_p$ is not a DM stack.

6.2 The Moduli Space $\mathcal{M}_g$

Theorem 6.6. The moduli space $\mathcal{M}_g$ is a DM stack (over $\mathbb{C}$) with $g \geq 2$.

Proof Idea. We will provide a relatively easy, constructive proof.

Claim: There is an isomorphism $\mathcal{M}_g \cong [U/G]$, where U is a smooth variety, $G = \text{PGL}(N)$, and the stabilizers are finite.

Indeed, proving this claim will also show that $\mathcal{M}_g$ is a smooth DM stack. We now proceed to the construction of the variety U and the group G .

Step 0: For any family of smooth projective curves of genus g , denoted by $\pi : \mathcal{C} \rightarrow S$, the morphism π is a projective map. That is, there exists a closed immersion $\mathcal{C} \hookrightarrow \mathbb{P}$ over S , where $\mathbb{P} \rightarrow S$ is a projective bundle.

Consider $\omega_{\mathcal{C}/S}$, the relative dualizing sheaf.

Fact: The sheaf $\omega_{\mathcal{C}/S}^{\otimes 3}$ is relatively very ample. (Recall that for any smooth curve C of genus $g(C) \geq 2$, the sheaf $\omega_C^{\otimes 3}$ is very ample).

Let $\mathbb{P} = \mathbb{P}(\pi_*(\omega_{\mathcal{C}/S}^{\otimes 3}))$. The dimension of the fibers is $h^0(C, \omega_C^{\otimes 3}) - 1$, which equals $5g - 6$ by the Riemann-Roch theorem. Thus, $\mathcal{C}$ is a closed subscheme of $\mathbb{P}$ over S via the canonical embedding given by $\omega_{\mathcal{C}/S}^{\otimes 3}$.

Step 1: Let H be the Hilbert scheme of closed subschemes in $\mathbb{P}^{5g-6}$ with the Hilbert polynomial $P(n) = \chi(\mathcal{O}_{\mathcal{C}}(n)) = (6n - 1)(g - 1)$, where $g(C) = g$. We define $U \subseteq H$ to be the locus parameterizing irreducible, smooth curves.

Fact: U is an open subset of H . and it is smooth³. (Note that the Hilbert scheme H itself is generally not smooth).

Step 2: Let $G = \mathrm{PGL}(5g - 5) = \mathrm{Aut}(\mathbb{P}^{5g-6})$, which acts naturally on $\mathbb{P}^{5g-6}$. We now verify the isomorphism $\mathcal{M}_g \cong [U/G]$.

Remark. Behind this isomorphism lies an important fact: for any smooth projective curves $C_1, C_2 \subseteq \mathbb{P}^{5g-6}$, $C_1 \cong C_2$ if and only if there exists an element $h \in G$ such that $h(C_1) = C_2$. (Hint for the proof: use the fact that any isomorphism must preserve the canonical bundles, hence $\omega_{C_1} \cong \omega_{C_2}$).

As a counterexample in other contexts, consider two degree 4 surfaces $S_1, S_2 \subseteq \mathbb{P}^3$. It is possible that $S_1 \cong S_2$ as abstract surfaces, but there does not exist any $h \in \mathrm{PGL}(4)$ such that $h(S_1) = S_2$.

Constructing the Functor: Let us construct the functor $F : \mathcal{M}_g \rightarrow [U/G]$. For any scheme $B \in \mathrm{Sch}_S$, we need to define a map $\mathcal{M}_g(B) \rightarrow [U/G](B)$.

An object in $\mathcal{M}_g(B)$ is a family of curves $\mathcal{C} \rightarrow B$. Under the functor F , this should map to an object in $[U/G](B)$, which consists of a principal G -bundle $P \rightarrow B$ along with a G -equivariant morphism $P \rightarrow U$.

What exactly is P ? By the definition of the quotient stack, $P \rightarrow B$ is a G -bundle, and $P \rightarrow U$ is G -equivariant. To understand this concretely, consider the case when the base B is just a point ($B = \mathrm{pt}$). In this case, the bundle is trivial, so $P = G$. A curve $\mathcal{C} \rightarrow \mathrm{pt}$ corresponds to a specific point $y \in U$. The G -equivariant map $P \rightarrow U$ is exactly the orbit map $G \rightarrow G \cdot y$. $\square$

We will give the opposite direction of the functor in the next lecture.

³Smoothness is left as exercise.

7 2026.4.1

7.1 Continuation of the moduli space $\mathcal{M}_g$

Theorem 7.1. $\mathcal{M}_g$ is a smooth Deligne-Mumford stack.

Theorem 7.2 (Deligne-Mumford). $\mathcal{M}_g$ is a separated smooth Deligne-Mumford stack.

Remark. *Separatedness means the following: suppose there exist families $\mathcal{C}_1 \rightarrow \Delta$ and $\mathcal{C}_2 \rightarrow \Delta$ over a “small neighborhood” Δ (for example, the spectrum of an Artinian local ring or the spectrum of a discrete valuation ring). If there is an isomorphism $\mathcal{C}_1|_{\Delta^*} \cong \mathcal{C}_2|_{\Delta^*}$ over the punctured neighborhood Δ^* , we can conclude that $\mathcal{C}_1 \cong \mathcal{C}_2$ over the entire Δ .*

This holds for families of polarized varieties. Suppose there exist families $(\mathcal{X}_i, \mathcal{L}_i) \rightarrow \Delta$ for $i = 1, 2$, where $\mathcal{L}_i$ is an ample line bundle (a polarization). If there is an isomorphism $\tilde{\varphi}$ between them over Δ^* :

$$\begin{array}{ccc} (\mathcal{X}_1, \mathcal{L}_1)|_{\Delta^*} & \xrightarrow[\sim]{\tilde{\varphi}} & (\mathcal{X}_2, \mathcal{L}_2)|_{\Delta^*} \\ \downarrow & & \downarrow \\ \Delta^* & \xlongequal{\quad} & \Delta^* \end{array}$$

Then $\tilde{\varphi}$ extends to an isomorphism over Δ . (The conclusion holds when considering curves.)

Continuation of the proof of the theorem: We want to show that $\mathcal{M}_g \cong [U/\mathrm{PGL}(5g-5)]$.

Consider the functor evaluated at a scheme B :

$$[U/\mathrm{PGL}(5g-5)](B) \longrightarrow \mathcal{M}_g(B)$$

An object in the groupoid on the left consists of a principal G -bundle $P \rightarrow B$ (where $G = \mathrm{PGL}(5g-5)$) together with a G -equivariant morphism $P \xrightarrow{\varphi} U$. We want to determine its image on the right side:

$$\left(\begin{array}{ccc} & P & \xrightarrow[\text{\scriptsize G-equiv.}]{\varphi} U \\ \text{\scriptsize principal G-bundle} \downarrow & & \\ & B & \end{array} \right) \longmapsto ?$$

Recall that $U \subset \mathrm{Hilb}(\mathbb{P}^{5g-6})$ parameterizes smooth curves in $\mathbb{P}^{5g-6}$ of genus g and degree $6g-6$.

Since the Hilbert scheme is representable, there exists a universal family $\mathcal{C} \rightarrow \mathrm{Hilb}$. By restriction, we obtain a family $\mathcal{C}_U \rightarrow U$, where $\mathcal{C}_U = \mathcal{C}|_U$.

The image “?” is given by taking the quotient of the pullback family:

$$? = \left((P \times_U \mathcal{C}_U \rightarrow P \rightarrow B) / G \right)$$

Here, the fibers of $P \times_U \mathcal{C}_U \rightarrow P$ are curves of genus g , and the fibers of $P \rightarrow B$ are isomorphic to G .

We omit the equivalence by showing that the functor is fully faithful and essentially surjective.

We want to show that the moduli stack $\mathcal{M}_g$ is smooth. There are two primary methods to establish this:

1. **Via Tangent Spaces:** Introduce the concept of the tangent space and the dimension of stacks. Smoothness is equivalent to the condition that the dimension of the tangent space is equal to the dimension of the stack at every point.
2. **Via Smooth Covers:** Show that there exists a surjective smooth cover $U \twoheadrightarrow \mathcal{M}_g$ in the smooth (or étale) topology, where U is a smooth scheme.

In either case, we need to calculate the dimension of some tangent space and the dimension of the corresponding stack or scheme.

- **For Method (1):** We need to compute $\dim \mathcal{M}_g$ and $\dim T_{[c]} \mathcal{M}_g$.
- **For Method (2):** We need to compute $\dim U$ (which is related to $\dim \text{Hilb}$, and we can consider this as known) and $\dim T_{[c]} U$.

To compute these tangent spaces, we will use deformation theory. The general structure of this approach is:

$$\text{Deformation theory} = \text{Deformation space} + \text{Obstruction space}$$

Here, the deformation space corresponds to the dimension of stack and the dimension of the tangent space.

7.2 Versal Spaces

Notations and Setup

Let Art_k denote the category of Artinian local rings. An object in this category is typically denoted by $(A, \mathfrak{m})$, where A is an Artinian local ring and $\mathfrak{m}$ is its maximal ideal, such that the residue field satisfies $A/\mathfrak{m} = k$.

Let $\mathcal{X}$ be a prestack over the category of schemes Sch_k . Suppose we are given a point $x_0 : \text{Spec}(k) \rightarrow \mathcal{X}$.

Definition 7.3 (Local deformation functor). The **local deformation functor** $\text{Def}_{\mathcal{X}, x_0} : \text{Art}_k \rightarrow \text{Set}$ is defined by mapping an Artinian local ring $(A, \mathfrak{m})$ to the set of isomorphism classes of deformations over A .

Definition 7.4 (Deformation of a point). A **deformation of the point x_0 over an Artinian local ring $(A, \mathfrak{m})$** is a pair $(x, \alpha) \in \mathcal{X}(A)$ such that its restriction $x|_k = x_0$.

Definition 7.5 (Prorepresentable functor). A functor $F : \text{Art}_k \rightarrow \text{Set}$ is said to be **prorepresentable** if there exists an isomorphism

$$F \cong \text{Mor}(R, -)|_{\text{Art}_k}$$

where $R \in \widehat{\text{Art}}_k$ is an object in the completion of the category of Artinian local rings. (For example $R = k[[\varepsilon]] \dots$)

Definition 7.6 (Universal Deformation). Let $\mathcal{X}$ be a prestack. If the local deformation functor $\text{Def}_{\mathcal{X}, x_0}$ is prorepresentable, the representing ring $R \in \widehat{\text{Art}}_k$ is called the **universal deformation ring** of x_0 .

A **formal deformation** is of the form $(R, \{x_n\})$ such that: For all $y \in \text{Def}_{\mathcal{X}, x_0}(A)$, there exists a unique morphism $\phi : R \rightarrow A$ such that $y \cong x \times_{\text{Spec } R} \text{Spec } A$, equivalently, there exists a unique morphism $R_n := R/\mathfrak{m}^{n+1} \rightarrow A$ such that $y \cong x_n \times_{\text{Spec } R_n} \text{Spec } A$ for an appropriate truncation level n .

Remark. $x_n \in \mathcal{X}(R_n)$, $x_n \times_{\text{Spec } R_n} \text{Spec } A$ is just the combination of $x_n : \text{Spec } R_n \rightarrow \mathcal{X}$ and the morphism $\text{Spec } A \rightarrow \text{Spec } R_n$.

Most stacks do not have universal deformation functors.

Definition 7.7 (Versal Deformation). A formal deformation $(R, \{x_n\})$ is **versal** if for all $y \in \text{Def}_{\mathcal{X}, x_0}(A)$, and any surjective morphism $A \twoheadrightarrow A_0$ in Art_k with $\mathfrak{m}_A^{m+1} = 0$, and any morphism $\phi_0 : R_m \rightarrow A_0$ such that there is an isomorphism $\alpha_0 : \phi_0^* x_m \xrightarrow{\sim} y|_{A_0}$, there exists a morphism $\phi : R_m \rightarrow A$ lifting $\phi_0 : R_m \rightarrow A_0$, and an isomorphism $\alpha : \phi^* x_m \xrightarrow{\sim} y$ extending α_0 .

Remark. It is called *miniversal* if "its dimension is minimal".

7.3 Examples

Example 7.8 (Deformation of Varieties). Let X be a smooth variety over a field k . We define the deformation functor

$$\text{Def}_X : \text{Art} \longrightarrow \text{Set}$$

from the category of local Artinian k -algebras to the category of sets. It maps an Artinian local ring $(A, \mathfrak{m})$ to the set of isomorphism classes of deformations of X over $\text{Spec}(A)$, which is given by:

$$(A, \mathfrak{m}) \longmapsto \left\{ \mathcal{X} \rightarrow \text{Spec}(A) \text{ such that } \mathcal{X} \times_{\text{Spec}(A)} \text{Spec}(A/\mathfrak{m}) \simeq X \right\} / \simeq$$

For a first-order deformation, we consider the specific case of the dual numbers, where $A = k[\varepsilon]/(\varepsilon^2)$. Notice that there is a standard short exact sequence associated with this ring:

$$0 \longrightarrow k \cdot \varepsilon \longrightarrow k[\varepsilon]/(\varepsilon^2) \longrightarrow k \longrightarrow 0$$

The space of first-order deformations evaluated at this ring is:

$$\text{Def}_X(k[\varepsilon]/(\varepsilon^2)) = \left\{ \mathcal{X} \rightarrow \text{Spec}(k[\varepsilon]/(\varepsilon^2)) \text{ such that } \mathcal{X} \times_{\text{Spec}(k[\varepsilon]/(\varepsilon^2))} \text{Spec}(k) \simeq X \right\} / \simeq$$

This set of isomorphism classes is naturally identified with the set of extensions of the structure sheaf $\mathcal{O}_X$ by itself. That is, it corresponds to the short exact sequences of sheaves of the form:

$$0 \longrightarrow \mathcal{O}_X \xrightarrow{\varepsilon} \mathcal{O}_{\mathcal{X}} \longrightarrow \mathcal{O}_X \longrightarrow 0$$

where the kernel is viewed as the ideal sheaf. This set of extensions is isomorphic to the first Ext group, which in turn is isomorphic to the first cohomology group of the tangent sheaf T_X :

$$\text{Def}_X(k[\varepsilon]/(\varepsilon^2)) \simeq \text{Ext}^1(\Omega_X^1, \mathcal{O}_X) \simeq H^1(X, T_X)$$

Example 7.9 (Smooth Projective Curve). If X is a smooth projective curve with genus $g \geq 2$, then by Serre duality, the dimension of the first-order deformation space is computed as follows:

$$\dim H^1(X, T_X) = \dim H^0(X, T_X^\vee \otimes \Omega_X)$$

Since the dual of the tangent bundle is the cotangent bundle ($T_X^\vee \simeq \Omega_X$), this becomes:

$$\dim H^1(X, T_X) = h^0(\Omega_X^{\otimes 2}) = 3g - 3$$

which use the condition that $g \geq 2$.

Example 7.10 (Quartic Surface in Projective Space). If $X \subseteq \mathbb{P}^3$ is a quartic surface, the dimension of its first-order deformation space is given by:

$$\dim \text{Def}_X(k[\varepsilon]/(\varepsilon^2)) = h^1(X, T_X) = h^1(X, \Omega_X) = 20$$

The details of this computation are left as an exercise.

Remark. To solve the exercise, one may use the fact that the canonical bundle of a quartic surface in $\mathbb{P}^3$ is trivial, meaning $K_X \simeq \mathcal{O}_X$. This implies that the tangent bundle is isomorphic to the cotangent bundle ($T_X \simeq \Omega_X$), and one can utilize the fact that $h^0(X, \Omega_X) = h^2(X, \Omega_X) = 0$.

Example 7.11 (Deformations of Bundles). Let X be a smooth curve and let E be a vector bundle on X . We define the deformation functor

$$\text{Def}_E : \text{Art} \longrightarrow \text{Set}$$

from the category of local Artinian k -algebras to the category of sets. It maps an Artinian local ring $(A, \mathfrak{m})$ to the set of isomorphism classes of vector bundles $\mathcal{E}$ on the product space $X \times_k \text{Spec}(A)$ such that its restriction to the central fiber is isomorphic to E . That is,

$$(A, \mathfrak{m}) \longmapsto \left\{ \mathcal{E} \text{ vector bundle on } X \times_k \text{Spec}(A) \mid \mathcal{E}|_{X \times_k \text{Spec}(k)} \simeq E \right\}$$

Example 7.12. Consider the case where $X = \mathbb{P}^1$ and the vector bundle is given by $E = \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$.

There exists a family of projective lines $\mathcal{X} \rightarrow B$ over a base B , and there exists a vector

bundle $\mathcal{E}$ on the total space $\mathcal{X}$ such that the restriction to a general fiber T_1 splits as:

$$\mathcal{E}|_{T_1} \simeq \mathcal{O}_{\mathbb{P}^1}(-1) \oplus \mathcal{O}_{\mathbb{P}^1}(-1)$$

while the restriction to the special fiber T_0 splits as:

$$\mathcal{E}|_{T_0} \simeq \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$$

Notice that throughout this deformation, both the degree and the rank of the bundle remain invariant.

The space of first-order deformations of the bundle E over the dual numbers is parametrized by the first Ext group, which is isomorphic to the first cohomology group of its endomorphism bundle:

$$\text{Def}_E(k[\varepsilon]/(\varepsilon^2)) \simeq \text{Ext}_{\mathcal{O}_X}^1(E, E) \simeq H^1(X, \mathcal{E}nd(E))$$

Remark. By analogy, this structure arises from considering the short exact sequence of vector bundles on the deformed space:

$$0 \longrightarrow E \longrightarrow \mathcal{E} \longrightarrow E \longrightarrow 0$$

Recall that previously, for the deformations of the variety itself, we considered the extension of sheaves:

$$0 \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_{\mathcal{X}} \longrightarrow \mathcal{O}_X \longrightarrow 0$$

which corresponded to elements in the group $\text{Ext}^1(\Omega_X, \mathcal{O}_X)$.

Remark. The direction of this deformation is strictly governed by the dimension of the first-order deformation space $H^1(X, \mathcal{E}nd(E))$:

- **Rigidity of the Generic Fiber:** If we start with $E = \mathcal{O}_{\mathbb{P}^1}(-1) \oplus \mathcal{O}_{\mathbb{P}^1}(-1)$, its endomorphism bundle is $\mathcal{E}nd(E) \simeq \mathcal{O}_{\mathbb{P}^1}^{\oplus 4}$. Since $H^1(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}^{\oplus 4}) = 0$, the first-order deformation space is trivial. This implies that $\mathcal{O}_{\mathbb{P}^1}(-1) \oplus \mathcal{O}_{\mathbb{P}^1}(-1)$ is **rigid** and cannot be deformed into any other splitting type.
- **Deformability of the Special Fiber:** Conversely, if we start with $E = \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$, its endomorphism bundle is $\mathcal{E}nd(E) \simeq \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(2) \oplus \mathcal{O}_{\mathbb{P}^1}(-2) \oplus \mathcal{O}_{\mathbb{P}^1}$. The dimension of the deformation space is given by $h^1(\mathbb{P}^1, \mathcal{E}nd(E)) = h^1(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(-2)) = 1$. This indicates that there exists a 1-dimensional family of non-trivial deformations.
- **Conclusion:** In algebraic geometry, deformations of vector bundles always flow towards a "more balanced" splitting type (by the upper semicontinuity of cohomology). Therefore,

the unbalanced bundle $\mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$ acts as the special fiber and deforms into the rigid generic fiber $\mathcal{O}_{\mathbb{P}^1}(-1) \oplus \mathcal{O}_{\mathbb{P}^1}(-1)$.

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We can categorize deformations into the following types:

- A **first-order deformation** is given by a family $\mathcal{X} \longrightarrow \operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$.
- A **formal deformation** is given by a family $\mathcal{X} \longrightarrow \operatorname{Spec} k[[t_1, \dots, t_n]]$.
- An **algebraic deformation** is given by a family $\mathcal{X} \longrightarrow \operatorname{Spec} k[S]$, where $k[S]$ represents a finitely generated k -algebra.

The general approach in deformation theory is to lift step-by-step: from a first-order deformation to a second-order deformation $\mathcal{X} \longrightarrow \operatorname{Spec} k[\varepsilon]/(\varepsilon^3)$, and continuing this process $\dots \rightsquigarrow \operatorname{Spec} k[[t]]$ to eventually obtain a formal deformation.

This process motivates the following fundamental questions:

1. **First-order deformations:** How do we compute the **space of first-order deformations**, denoted as $\operatorname{Def}_{1\text{st}}(X_0) := \operatorname{Def}_{X_0}(\operatorname{Spec} k[\varepsilon]/(\varepsilon^2))$? For example, if X_0 is a scheme or a manifold, this space is given by the first cohomology group of the tangent bundle, that is, $\operatorname{Def}_{1\text{st}}(X_0) = H^1(X_0, T_{X_0})$.
2. **Lifting to higher-order deformations:** Can we lift a given deformation to a higher order? In general, there exist certain obstructions to this lifting process.

Remark. *In many cases, if the obstruction of the first-order deformation is trivial, then there are no obstructions to extending it to a full formal deformation.*

3. **Algebraicization:** Once a formal deformation is obtained, one must check whether this formal deformation is algebraic (i.e., whether it can be algebraized).

Example 8.1. Consider the deformations of a smooth variety X .

1. The space of first-order deformations is given by $\operatorname{Def}_{1\text{st}}(X) = H^1(X, T_X)$, which is also the **tangent space of the versal deformation space**, denoted as $T_{\operatorname{Def}(X)}$.
2. We consider the obstructions to lifting. For every first-order deformation $\mathcal{X}_1 \longrightarrow \operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$ of the variety $X \longrightarrow \operatorname{Spec} k$, i.e.:

$$\mathcal{X}_1 \times_{\operatorname{Spec} k[\varepsilon]/(\varepsilon^2)} \operatorname{Spec} k \cong X,$$

we ask if it can be lifted to a second-order deformation $\mathcal{X}_2 \longrightarrow \operatorname{Spec} k[\varepsilon]/(\varepsilon^3)$.

3. **Algebraicization:** We consider the process of algebraicizing a formal deformation

$\mathcal{X} \longrightarrow \operatorname{Spec} k[[t]]$ to an algebraic deformation $\mathcal{X} \longrightarrow \operatorname{Spec} k[S]$.

The obstruction space for this lifting problem is given by $\operatorname{Obs}(X) \cong H^2(X, T_X)$.

Remark. *Regarding the obstruction space: there exists an obstruction to lifting only if there exists a non-trivial cohomology class. In particular, if the obstruction space $\operatorname{Obs}(X) = 0$, then there exist no obstructions to lifting.*

Remark. *If there exists no obstruction, we may call the deformation space $\operatorname{Def}(X)$ smooth. (Regarding the dimension of the obstruction space, we have the following inequality:*

$$\dim \operatorname{Obs}(X) \geq \dim T_{\operatorname{Def}(X)} - \dim \operatorname{Def}(X).$$

Recall that the tangent space of the variety, T_X , is given by the set of morphisms

$$\{\operatorname{Spec} k[\varepsilon]/(\varepsilon^2) \longrightarrow X\},$$

while the tangent space of the deformation space, $T_{\operatorname{Def}(X)}$, is given by the set of first-order deformations

$$\{\mathcal{X} \longrightarrow \operatorname{Spec} k[\varepsilon]/(\varepsilon^2)\}.$$

Theorem 8.2 (Grothendieck's Existence Theorem). *If the formal scheme $\mathcal{X}$ admits a polarization (which is equivalent to being projective), then this algebraicization holds.*

8.1 Embedded Deformations

Let X be a smooth variety. Suppose $X \subseteq \mathbb{P}^N$, we consider the deformations of X embedded in $\mathbb{P}^N$. A deformation of the embedding $X \subseteq \mathbb{P}^N$ over a base scheme $\operatorname{Spec} B$ is given by a family of the following form:

$$\begin{array}{ccc} \mathcal{X} & \hookrightarrow & \mathbb{P}_B^N \\ \downarrow & & \\ \operatorname{Spec} B & & \end{array}$$

The space of first-order embedded deformations is a subspace of the general first-order deformations of X , that is:

$$\operatorname{Def}_{1\text{st}}(X \subseteq \mathbb{P}^N) \subseteq H^1(X, T_X).$$

Theorem 8.3. *We have the following conclusion for the space of first-order embedded*

deformations:

$$\mathrm{Def}_{\mathrm{1st}}(X \subseteq \mathbb{P}^N) \cong H^0(X, \mathcal{N}_X),$$

where $\mathcal{N}_X$ is the normal bundle of X in $\mathbb{P}^N$.

Remark. *Intuitively, this isomorphism indicates that deforming X inside $\mathbb{P}^N$ corresponds to moving X along the direction of its normal vectors.*

If $X \subseteq \mathbb{P}^N$ is a smooth hypersurface, how do we compute $\mathrm{Def}_{\mathrm{1st}}(X \subseteq \mathbb{P}^N)$?

Consider a smooth hypersurface $X \subseteq \mathbb{P}^N$ defined by a homogeneous polynomial equation $f(x) = 0$, where $x = [x_0 : \cdots : x_N]$ are the homogeneous coordinates. Let $d = \deg X = \deg f$.

The space of first-order embedded deformations of X in $\mathbb{P}^N$ (which corresponds to deformations over the dual numbers $k[\varepsilon]/(\varepsilon^2)$) can be explicitly described as the set of perturbed polynomial equations:

$$\mathrm{Def}_{\mathrm{1st}}(X \subseteq \mathbb{P}^N) = \left\{ \{F(x, \varepsilon) = 0\} \subseteq \mathbb{P}_{k[\varepsilon]/(\varepsilon^2)}^N \mid \deg F = \deg f, \text{ and } F(x, 0) = f(x) \right\}.$$

For $H^0(X, \mathcal{N}_X)$, the normal bundle is given by $\mathcal{N}_X \cong \mathcal{O}_X(d)$. Consequently, we can compute this space by the exact sequence $0 \rightarrow \mathcal{O}_{\mathbb{P}^N} \rightarrow \mathcal{O}_{\mathbb{P}^N}(d) \rightarrow \mathcal{N}_X \rightarrow 0$ as:

$$H^0(X, \mathcal{N}_X) \cong H^0(\mathbb{P}^N, \mathcal{O}_{\mathbb{P}^N}(d)) / (k \cdot f)$$

By considering $[g(x)] \mapsto \{f(x) + \varepsilon g(x) = 0\}$, It is easy to see the equivalence between the two descriptions of the space of first-order embedded deformations.

Let $Y \subseteq \mathbb{P}^4$ be a smooth hypersurface defined by the equation $Y = \{F(x_0, \dots, x_4) = 0\}$. Let $X \subseteq Y$ be a smooth rational curve, which implies $X \simeq \mathbb{P}^1$.

We want to compute the space of first-order embedded deformations of X inside Y , denoted as $\mathrm{Def}_{\mathrm{1st}}(X \subseteq Y)$. The conclusion is that this space is given by the global sections of the normal bundle, $H^0(X, \mathcal{N}_{X/Y})$.

By definition, this space of first-order deformations corresponds to families of the form:

$$\mathcal{X} \subseteq Y \times \mathrm{Spec} k[\varepsilon]/(\varepsilon^2)$$

that are flat over the base $\mathrm{Spec} k[\varepsilon]/(\varepsilon^2)$.

Since $X \simeq \mathbb{P}^1$, we can explicitly parameterize the original curve X using coordinates $t \in \mathbb{P}^1$:

$$X = \{[a_0(t) : \cdots : a_4(t)] \mid t \in \mathbb{P}^1\},$$

where $a_i(t) \in \mathbb{C}[t]$ are polynomials of degree $\deg(a_i(t)) = m$. Because the curve X is contained within the hypersurface Y , these parameterized coordinates must satisfy the defining equation of Y for all t :

$$F(a_0(t), \dots, a_4(t)) \equiv 0 \quad \text{for all } t.$$

A first-order deformation $\mathcal{X}$ can then be represented by perturbing these parameterizing polynomials over the dual numbers:

$$\mathcal{X} = \{[a_0(t, \varepsilon) : \dots : a_4(t, \varepsilon)]\},$$

subject to the conditions that the reduction modulo ε recovers the original curve, meaning $a_i(t, 0) = a_i(t)$, and that the degree with respect to t remains constant, meaning $\deg_t a_i(t, \varepsilon) = m$.

Question: How can we explicitly see that this parameterized deformation space is related to the cohomology group $H^0(X, \mathcal{N}_{X/Y})$?

For any section $\sigma \in H^0(X, \mathcal{N}_{X/Y})$, we can represent it by an array of polynomials $\sigma = (\sigma_0(t), \dots, \sigma_4(t))$ of degree m ⁴ that satisfies the following condition:

$$\sum_{i=0}^4 \sigma_i(t) \cdot \frac{\partial F}{\partial x_i}(a(t)) = 0.$$

This equation ensures that the deformation vectors geometrically lie within the tangent space of Y , meaning it defines a section of $T_Y|_X$.

Furthermore, such a section represents the trivial element $\sigma = 0 \in H^0(X, \mathcal{N}_{X/Y})$ if and only if it is induced by a vector field on X itself. That is, it can be expressed as:

$$\sigma_i(t) = \alpha \cdot \frac{\partial a_i}{\partial t_0} + \beta \cdot \frac{\partial a_i}{\partial t_1},$$

where $t = [t_0 : t_1]$ are the homogeneous coordinates of $X \simeq \mathbb{P}^1$ and α, β are of degree 1.

We can formalize this computation using the standard short exact sequence of tangent and normal bundles:

$$0 \longrightarrow T_X \longrightarrow T_Y|_X \longrightarrow \mathcal{N}_{X/Y} \longrightarrow 0.$$

Now we consider the **obstructions** to lifting these deformations.

For any first-order deformation $\sigma(t) \in \text{Def}_{\text{1st}}(X \subseteq Y)$, we require the degree to be preserved, meaning $\deg \sigma_i(t) = \deg a_i(t) = m$. The condition that the deformed curve remains

⁴We have $0 \rightarrow \mathcal{O}_{\mathbb{P}^4} \rightarrow \mathcal{O}_{\mathbb{P}^4}(1)^{\oplus 5} \rightarrow T \rightarrow 0$, pulling back along $\mathbb{P}^1 \rightarrow \mathbb{P}^4$, $\iota^* \mathcal{O}_{\mathbb{P}^4}(1) = \mathcal{O}_{\mathbb{P}^1}(m)$.

inside Y implies it must satisfy the defining equation over the dual numbers $k[\varepsilon]/(\varepsilon^2)$:

$$F(a(t, \varepsilon)) = 0, \quad \text{where } \varepsilon \in k[\varepsilon]/(\varepsilon^2).$$

By substituting the expansion $a_i(t, \varepsilon) = a_i(t) + \varepsilon \sigma_i(t)$ into the equation, we obtain:

$$F(a_0(t) + \varepsilon \sigma_0(t), \dots, a_4(t) + \varepsilon \sigma_4(t)) = 0.$$

Taking the derivative with respect to ε , we recover the exact linear equation discussed previously. This confirms our characterization of the first-order deformations.

Remark. We try to compute $\dim H^0(X, \mathcal{N}_{X/Y})$ by:

$$\dim H^0(X, \mathcal{N}_{X/Y}) = \dim H^0(X, T_Y|_X) - \dim H^0(X, T_X)$$

recall $\deg a_i = m$ and $\deg F = N$, F and a_i are fixed.

1. **Total Parameters:** The sections $\sigma_i(t)$ are 5 polynomials of degree m . The dimension of this parameter space is $5(m+1)$.
2. **Tangent Space Constraints:** The condition $\sum \sigma_i(t) \frac{\partial F}{\partial x_i}(a(t)) \equiv 0$ is a polynomial identity. Since $\deg \frac{\partial F}{\partial x_i} = N-1$, the evaluated term $\frac{\partial F}{\partial x_i}(a(t))$ has degree $(N-1)m$. Multiplying by $\sigma_i(t)$ of degree m results in a polynomial of total degree $(N-1)m+m = Nm$. Setting a polynomial of degree Nm to zero imposes $Nm+1$ linear conditions.
3. **Quotient by Automorphisms:** There are two sources of redundancy:
 - Tangent vectors coming from X itself, given by $H^0(X, T_X)$. For $X \simeq \mathbb{P}^1$, we have $\dim H^0(\mathbb{P}^1, T_{\mathbb{P}^1}) = 3$.
 - The trivial radial scaling vector field from the Euler sequence of $\mathbb{P}^4$ (i.e., $\sigma_i(t) = c \cdot a_i(t)$), which contributes 1 dimension.

$$\text{Therefore, } \dim H^0(X, \mathcal{N}_{X/Y}) = [5(m+1) - (Nm+1) - 1] - 3.$$

Substituting these into the formula with $N = 5$:

$$\begin{aligned} \text{Expected } \dim H^0(X, \mathcal{N}_{X/Y}) &= 5(m+1) - (5m+1) - 4 \\ &= 5m+5 - 5m-1-4 \\ &= 0. \end{aligned}$$

This implies that rational curves on a generic quintic threefold ($N = 5$ in $\mathbb{P}^4$) have an expected deformation dimension of 0, meaning they are infinitesimally rigid.

The fundamental question then becomes: can we **lift this to the second order** (and subsequently to higher orders)?

To investigate this, we attempt to solve the equation $F(a(t, \varepsilon)) = 0$ formally, seeking a solution in the formal power series ring $k[[\varepsilon]]$:

$$a_i(t, \varepsilon) = a_i(t) + \varepsilon \sigma_i(t) + \frac{\varepsilon^2}{2} \sigma_i^{(2)}(t) + \cdots \in k[[\varepsilon]].$$

Suppose there exists such a solution up to order $n - 1$. We substitute this series into $F(a(t, \varepsilon)) = 0$. By taking the n -th differential with respect to ε , we isolate the n -th order terms and obtain an equation of the following form:

$$\sum_{i=0}^4 \frac{\partial F}{\partial x_i}(a(t)) \sigma_i^{(n)}(t) = \text{lower-order terms}.$$

Observe that the right-hand side is a known expression consisting exclusively of terms involving lower-order coefficients $\sigma_i^{(k)}(t)$ for $k < n$; it contains no terms involving the n -th order unknown $\sigma_i^{(n)}(t)$.

Remark. *This formulation yields a system of linear equations for the unknown n -th order components $\sigma_i^{(n)}(t)$. When computing $\sum_{i=0}^4 \frac{\partial F}{\partial x_i}(a(t)) \sigma_i^{(n)}(t)$, for every l , you should matching $\left(\frac{\partial F}{\partial x_i}(a(t))\right)_{\deg_t=j}$ and $(\sigma_i^{(n)}(t))_{\deg_t=l-j}$ to get the coefficient of t^l in $\sum_{i=0}^4 \frac{\partial F}{\partial x_i}(a(t)) \sigma_i^{(n)}(t)$, which you can get a coefficient matrix. There exists a solution to this system if and only if the coefficient matrix has full rank.*

Theorem 8.4 (Artin's Criterion). Let $\mathcal{X}$ be a prestack over the category of schemes (Sch). The prestack $\mathcal{X}$ is an algebraic stack if and only if the following conditions hold:

1. The diagonal morphism $\Delta : \mathcal{X} \longrightarrow \mathcal{X} \times \mathcal{X}$ is representable by algebraic spaces.
2. $\mathcal{X}$ is limit preserving.
3. For every object $X_0 \in \mathcal{X}(\text{Spec } k)$, there exists a formal versal deformation.
4. Every formal object of $\mathcal{X}$ is effective.
5. $\mathcal{X}$ satisfies the openness of versality.

Remark. *The overarching philosophy of Artin's Criterion is to establish an equivalence between local formal properties and global algebraic properties (local $\iff$ global).*

Specifically, conditions (3), (4), and (5) collectively serve to guarantee the existence of a smooth cover for the stack $\mathcal{X}$. The intuitive idea for constructing this cover is as follows:

- *We start by deforming a point. Condition (3) guarantees the existence of a formal deformation.*
- *Condition (5) ensures that this formal versality is an open condition, which allows us to extend the property to an open neighborhood, providing local smoothness.*
- *Finally, condition (2) guarantees that these formal objects are effective, meaning they can be algebraicized and glued together to form the desired global smooth cover.*

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Let X be a smooth projective variety over a field k , where $k = \bar{k}$ and $\text{char}(k) = 0$ (for example, we may take $k = \mathbb{C}$). Let $\dim X = d$, and let $\mathcal{H} = \mathcal{O}_X(1)$ be an ample line bundle on X .

Goal: The main objective is to construct a moduli space of semistable torsion-free sheaves on X with fixed topological invariants.

9.1 Stack of Coherent Sheaves

Definition 9.1 (Stack of Coherent Sheaves). The **stack of coherent sheaves**, denoted by $\mathcal{Coh}_X$, is a functor from the category of schemes over k to the category of groupoids:

$$\begin{aligned} \mathcal{Coh}_X : \text{Sch}_k &\longrightarrow \text{Groupoids} \\ T &\longmapsto \{\text{coherent sheaves on } X \times T \text{ that are flat over } T\}. \end{aligned}$$

Remark. The condition of being flat over the base T captures the intuitive notion of “continuity” in algebraic geometry.

Theorem 9.2. The stack of coherent sheaves $\mathcal{Coh}_X$ is an Artin stack, and it is locally of finite type.

Proof Sketch. We outline the proof in the following steps:

1. By the property of étale descent for coherent sheaves, $\mathcal{Coh}_X$ satisfies the stack condition, which implies that $\mathcal{Coh}_X$ is indeed a stack.
2. We acknowledge the existence of the Quot functor. Define the functor $\text{Quot}_{N,X} : \text{Sch}_k \rightarrow \text{Sets}$ by

$$T \longmapsto \left\{ \text{quotients } \mathcal{O}_{X \times T}^{\oplus N} \twoheadrightarrow \mathcal{F} \mid \mathcal{F} \text{ is flat over } T \right\} / \sim.$$

Fact: It is a foundational result that $\text{Quot}_{N,X}$ is represented by a scheme.

3. We can construct morphisms $p_{N,n} : \text{Quot}_{N,X} \longrightarrow \mathcal{Coh}_X$ mapping a given quotient $(\mathcal{O}_{X \times T}^{\oplus N} \twoheadrightarrow \mathcal{F})$ to the twisted sheaf $\mathcal{F}(-n)$.

The rationale behind this is as follows: a surjection $\mathcal{O}^{\oplus N} \twoheadrightarrow \mathcal{F}$ implies that $\mathcal{F}$ is globally generated. However, an arbitrary coherent sheaf is not necessarily globally generated. To resolve this, we use Serre’s theorem, which guarantees that the twisted sheaf $\mathcal{F}(n)$ is globally generated for some sufficiently large integer n .

Taking the disjoint union over all N and n , we obtain a morphism

$$p = \coprod_{N,n} p_{N,n} : \coprod_{N,n} \mathrm{Quot}_{N,X} \longrightarrow \mathrm{Coh}_X.$$

4. We define an open subscheme $\mathrm{Quot}_{N,X}^\circ \subset \mathrm{Quot}_{N,X}$ to parametrize objects with a suitably uniform structure (parametrizing framed objects). Specifically, it is defined by considering the projection morphism $p : X \times T \rightarrow T$ together with the quotient map $\alpha : \mathcal{O}_{X \times T}^{\oplus N} \twoheadrightarrow \mathcal{F}$, such that the following two conditions hold:
 - (a) The higher direct image sheaves vanish, meaning $R^i p_* \mathcal{F} = 0$ for all $i > 0$.
 - (b) The naturally induced map on the pushforwards $p_* \mathcal{O}_{X \times T}^{\oplus N} = \mathcal{O}_T^{\oplus N} \rightarrow p_* \mathcal{F}$ is an isomorphism.
5. Finally, we consider the restricted morphism $p^\circ : \mathrm{Quot}_{N,X}^\circ \rightarrow \mathrm{Coh}_X$. This morphism is smooth. To see why this is the case, we can observe that locally over the base scheme T , the fiber of this morphism is given by $\mathrm{Isom}(\mathcal{O}_T^{\oplus N}, p_* \mathcal{F})$. This space of isomorphisms admits a natural action by the general linear group GL_N , essentially making the fiber a GL_N -torsor.

□

Remark. The quotient stack $[\mathrm{Quot}/\mathrm{GL}_N]$ can be approximated by the Geometric Invariant Theory quotient $\mathrm{Quot}/\mathrm{GL}_N$, which provides a coarse moduli space.

Remark. There are two important aspects of the Quot scheme:

1. The scheme Quot generally has infinitely many connected components. To reduce this to a finite number of components (or to focus on a specific one), we must fix the numerical invariants of the sheaves. This means fixing the rank $\mathrm{rank}(\mathcal{F})$ and the Chern classes $c_1(\mathcal{F}), c_2(\mathcal{F})$, and so on.

Equivalently, this is the same as fixing the Chern character:

$$\mathrm{ch}(\mathcal{F}) = \mathrm{rank}(\mathcal{F}) + c_1(\mathcal{F}) + \frac{c_1(\mathcal{F})^2 - 2c_2(\mathcal{F})}{2} + \dots$$

By the Hirzebruch-Riemann-Roch theorem, fixing these invariants is equivalent to fixing the Hilbert polynomial:

$$p(\mathcal{F}, m) = \chi(\mathcal{F} \otimes \mathcal{O}(m)).$$

2. The scheme Quot has the boundedness property, which we will give the definition later.

Essentially, merely fixing the numerical invariants is not sufficient to guarantee that the family of sheaves is bounded.

Example 9.3. Let $X = \mathbb{P}^1$. Consider the collection of sheaves $\{\mathcal{O}(-n) \oplus \mathcal{O}(n)\}_{n \geq 0}$. Notice that for all $n \geq 0$, these sheaves share the same numerical invariants (rank is 2 and the first Chern class $c_1 = 0$). We claim that there is no family parameterized by a scheme of finite type that contains all of these sheaves.

Proof. Suppose for the sake of contradiction that there exists a scheme S of finite type over k , and a universal family $\mathcal{E} \in \mathcal{Coh}(X \times S)$ containing all of them. This implies that for every $i \geq 0$, there exists a point $s_i \in S$ such that the fiber $\mathcal{E}_{s_i}$ satisfies:

$$\mathcal{E}_{s_i} \cong \mathcal{O}(-i) \oplus \mathcal{O}(i).$$

Consider the function measuring the dimension of global sections:

$$\begin{aligned} h^0: S &\longrightarrow \mathbb{Z}_{\geq 0} \\ s &\longmapsto h^0(\mathbb{P}^1, \mathcal{E}_s). \end{aligned}$$

By the semicontinuity theorem, this function is upper semi-continuous. Consequently, for any integer n , the locus $S_{\geq n} = \{s \in S \mid h^0(\mathbb{P}^1, \mathcal{E}_s) \geq n\}$ is a closed subset of S .

Since S is a scheme of finite type, the function h^0 must be globally bounded on S . Therefore, we must have $S_{\geq m} = \emptyset$ for some sufficiently large integer $m \gg 0$.

However, by direct computation for our sequence of sheaves, we have:

$$h^0(\mathbb{P}^1, \mathcal{O}(-n) \oplus \mathcal{O}(n)) = h^0(\mathbb{P}^1, \mathcal{O}(n)) = n + 1.$$

This means the dimension of global sections grows without bound as n increases, which contradicts the fact that $S_{\geq m} = \emptyset$ for large m . Thus, no such scheme S of finite type can exist. $\square$

9.2 Boundedness and Regularities

Definition 9.4 (bounded family of sheaves). A set of coherent sheaves $\{\mathcal{F}_\lambda\}_{\lambda \in \Lambda}$ on a scheme X is called **bounded** if there exists a scheme S of finite type and a coherent sheaf $\mathcal{F} \in \mathcal{Coh}(X \times S)$ such that for every $\lambda \in \Lambda$, we have $\mathcal{F}_\lambda \cong \mathcal{F}_s$ for some closed point $s \in S$.

Definition 9.5 (m -regular sheaf). A coherent sheaf $\mathcal{F} \in \text{Coh}(X)$ is said to be **m -regular** (in the sense of Castelnuovo-Mumford) if

$$H^i(X, \mathcal{F}(m-i)) = 0 \quad \text{for all } i > 0.$$

Lemma 9.6. If a coherent sheaf $\mathcal{F}$ is m -regular, then:

- (1) $\mathcal{F}$ is m' -regular for all $m' \geq m$.
- (2) $\mathcal{F}(m)$ is globally generated.

Proof of (1). We prove part (1). Let $H \subset X$ be a general hyperplane section. We have the standard short exact sequence:

$$0 \longrightarrow \mathcal{F}(-1) \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}|_H \longrightarrow 0 \quad (*)$$

Tensoring the sequence $(*)$ with $\mathcal{O}_X(m-i)$, we obtain the following segment of the long exact sequence in cohomology:

$$\cdots \longrightarrow H^i(X, \mathcal{F}(m-i)) \longrightarrow H^i(H, \mathcal{F}|_H(m-i)) \longrightarrow H^{i+1}(X, \mathcal{F}(m-i-1)) \longrightarrow \cdots$$

Since $\mathcal{F}$ is m -regular, we know that $H^i(X, \mathcal{F}(m-i)) = 0$ for all $i > 0$. Furthermore, substituting $i+1$ into the m -regularity condition gives $H^{i+1}(X, \mathcal{F}(m-(i+1))) = 0$. Therefore, both the left and right terms in the exact sequence above vanish, which implies that

$$H^i(H, \mathcal{F}|_H(m-i)) = 0 \quad \text{for all } i > 0.$$

This shows that $\mathcal{F}|_H$ is m -regular on H .

We now proceed by induction on the dimension of X :

- **Base case:** If $\dim X = 0$, statement (1) holds naturally, as there are no higher cohomology groups for $i > 0$.
- **Inductive step:** Assume that statement (1) holds for varieties of dimension $\leq \dim X - 1$. Since $\dim H = \dim X - 1$ and $\mathcal{F}|_H$ is m -regular, the induction hypothesis implies that $\mathcal{F}|_H$ is m' -regular for all $m' \geq m$.

Next, tensor the exact sequence $(*)$ with $\mathcal{O}_X(m-i+1)$. The associated long exact cohomology sequence yields:

$$\cdots \longrightarrow H^i(X, \mathcal{F}(m-i)) \longrightarrow H^i(X, \mathcal{F}(m-i+1)) \longrightarrow H^i(H, \mathcal{F}|_H(m-i+1)) \longrightarrow \cdots$$

Here, the first term is $H^i(X, \mathcal{F}(m-i)) = 0$ because $\mathcal{F}$ is m -regular. The third term is $H^i(H, \mathcal{F}|_H(m-i+1)) = 0$ because $\mathcal{F}|_H$ is $(m+1)$ -regular (by the induction hypothesis applied to $\mathcal{F}|_H$).

Consequently, the middle term must also vanish:

$$H^i(X, \mathcal{F}(m-i+1)) = 0 \quad \text{for all } i > 0.$$

This proves that $\mathcal{F}$ is $(m+1)$ -regular. By iterating this argument, we conclude that $\mathcal{F}$ is m' -regular for all $m' \geq m$. This completes the proof of part (1). $\square$

Proof of (2). Let $V = H^0(X, \mathcal{O}(1))$. We consider the Koszul complex associated to V :

$$\cdots \longrightarrow \bigwedge^2 V \otimes \mathcal{O}_X(-2) \longrightarrow V \otimes \mathcal{O}_X(-1) \longrightarrow \mathcal{O}_X \longrightarrow 0.$$

We tensor this complex with $\mathcal{F}(m+k)$. There is a fact that if we break the Koszul complex into short exact sequences, tensoring each of these short exact sequences with $\mathcal{F}(m+k)$ preserves exactness. ⁵

By analyzing the long exact sequences in cohomology associated with these short exact sequences, we encounter terms of the form $H^i(X, \bigwedge^i V \otimes \mathcal{F}(m-i+k))$. Since $\mathcal{F}$ is m -regular, we know from Part (1) that $\mathcal{F}$ is also $(m+k)$ -regular for any $k \geq 0$. This regularity condition precisely implies that the higher cohomology groups vanish:

$$H^i \left(X, \bigwedge^i V \otimes \mathcal{F}(m+k-i) \right) = 0 \quad \text{for all } i > 0.$$

Because of this vanishing, the map on global sections induced by the Koszul complex is surjective. In particular, we have a surjection:

$$H^0(X, \mathcal{F}(m+k)) \otimes H^0(X, \mathcal{O}(1)) \cong H^0(X, V \otimes \mathcal{F}(m+k)) \twoheadrightarrow H^0(X, \mathcal{F}(m+k+1)).$$

By iterating this surjection (induction on k), we obtain a surjective map for any $k \geq 0$:

$$H^0(X, \mathcal{F}(m)) \otimes H^0(X, \mathcal{O}(k)) \twoheadrightarrow H^0(X, \mathcal{F}(m+k)). \quad \text{— (1)}$$

Now, by Serre's Theorem, for an integer $k \gg 0$ sufficiently large, the twisted sheaf $\mathcal{F}(m+k)$

⁵We first have $0 \rightarrow Z_0 \rightarrow V \otimes \mathcal{O}_X(-1) \rightarrow \mathcal{O}_X \rightarrow 0$. Both $\mathcal{O}_X$ and $V \otimes \mathcal{O}_X(1)$ are locally free, so is Z_0 . Similarly we have $Z_i := \ker \left(\bigwedge^{i+1} V \otimes \mathcal{O}_X(-i-1) \rightarrow \bigwedge^i V \otimes \mathcal{O}_X(-i) \right)$ are locally free. We also know that $\text{Tor}_1^{\mathcal{O}_X}(\mathcal{E}, \mathcal{F}(m+k)) = 0$ with $\mathcal{E}$ locally free. So tensoring with $\mathcal{F}(m+k)$ preserves exactness.

is globally generated. This means we have a natural surjection:

$$H^0(X, \mathcal{F}(m+k)) \otimes \mathcal{O}_X \twoheadrightarrow \mathcal{F}(m+k).$$

Tensoring this entire surjection with $\mathcal{O}_X(-k)$, we obtain:

$$H^0(X, \mathcal{F}(m+k)) \otimes \mathcal{O}_X(-k) \twoheadrightarrow \mathcal{F}(m).$$

Finally, we combine this with the surjectivity established in (1). Tensoring the surjective map (1) with $\mathcal{O}_X(-k)$, we get a sequence of surjective morphisms:

$$H^0(X, \mathcal{F}(m)) \otimes (H^0(X, \mathcal{O}(k)) \otimes \mathcal{O}_X(-k)) \twoheadrightarrow H^0(X, \mathcal{F}(m+k)) \otimes \mathcal{O}_X(-k) \twoheadrightarrow \mathcal{F}(m).$$

The composition of these maps is surjective. Notice that the left-hand side factors through the natural evaluation map $H^0(X, \mathcal{O}(k)) \otimes \mathcal{O}_X(-k) \rightarrow \mathcal{O}_X$. Therefore, the natural evaluation map for $\mathcal{F}(m)$ itself:

$$H^0(X, \mathcal{F}(m)) \otimes \mathcal{O}_X \longrightarrow \mathcal{F}(m)$$

must also be surjective. This concludes the proof that $\mathcal{F}(m)$ is globally generated. $\square$

Theorem 9.7 (Mumford). A set of coherent sheaves $\{\mathcal{F}_\lambda\}$ is bounded if and only if the set of their Hilbert polynomials $\{p(\mathcal{F}_\lambda)\}$ is finite, and the set of their Castelnuovo-Mumford regularities $\{\rho(\mathcal{F}_\lambda)\}$ has an upper bound.

(Here, the **Castelnuovo-Mumford regularity** is defined as $\rho(\mathcal{F}) = \min\{m \mid \mathcal{F} \text{ is } m\text{-regular}\}$.)

Definition 9.8. Let $\mathcal{F} \in \text{Coh}(X)$ be a coherent sheaf.

- (1) A hyperplane section $H \in |\mathcal{O}_X(1)|$ is called **$\mathcal{F}$ -regular** if the following sequence is exact:

$$0 \longrightarrow \mathcal{F}(-1) \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}|_H \longrightarrow 0.$$

- (2) A sequence of hyperplane sections $H_1, \dots, H_l \in |\mathcal{O}_X(1)|$ is called an **$\mathcal{F}$ -regular sequence** if, for each index i , the hyperplane H_i is $\mathcal{F}|_{H_1 \cap \dots \cap H_{i-1}}$ -regular.

Theorem 9.9 (Kleiman). Let $\{\mathcal{F}_\lambda\}_{\lambda \in \Lambda}$ be a set of coherent sheaves with a fixed Hilbert polynomial, meaning $p(\mathcal{F}_\lambda) = P$ for all $\lambda \in \Lambda$.

The set $\{\mathcal{F}_\lambda\}$ is bounded if and only if there exist constants C_j for $0 \leq j \leq \dim X$ such

that for every $\lambda \in \Lambda$, there exists an $\mathcal{F}_\lambda$ -regular sequence $H_1, \dots, H_d \in |\mathcal{O}_X(1)|$ satisfying

$$h^0(X, \mathcal{F}_\lambda|_{H_1 \cap \dots \cap H_l}) \leq C_l \quad \text{for all } 0 \leq l \leq d.$$

9.3 Stabilities

Definition 9.10 (Slope Stability). Let $(X, \mathcal{H})$ be a projective variety with a fixed ample line bundle $\mathcal{H}$, and let $\mathcal{E} \in \mathcal{Coh}(X)$ be a torsion-free sheaf. The **slope** of $\mathcal{E}$ is defined as

$$\mu_{\mathcal{H}}(\mathcal{E}) = \frac{c_1(\mathcal{E}) \cdot \mathcal{H}^{d-1}}{\text{rk}(\mathcal{E})}$$

where $d = \dim \text{Supp}(\mathcal{E})$.

A torsion-free sheaf $\mathcal{E}$ is called **$\mu_{\mathcal{H}}$ -stable** (respectively, **$\mu_{\mathcal{H}}$ -semistable**) if for all proper subsheaves $0 \subsetneq \mathcal{F} \subsetneq \mathcal{E}$ with $\text{rk}(\mathcal{F}) < \text{rk}(\mathcal{E})$, we have

$$\mu_{\mathcal{H}}(\mathcal{F}) < \mu_{\mathcal{H}}(\mathcal{E}) \quad (\text{respectively, } \mu_{\mathcal{H}}(\mathcal{F}) \leq \mu_{\mathcal{H}}(\mathcal{E})).$$

Definition 9.11 (Gieseker Stability). Consider the Hilbert polynomial of $\mathcal{E}$:

$$P(\mathcal{E}, m) = \chi(X, \mathcal{E} \otimes \mathcal{O}(m\mathcal{H})) = \sum_{i=0}^d \alpha_i(\mathcal{E}) \frac{m^i}{i!}$$

The leading coefficients of this polynomial are given by:

$$\begin{aligned} \alpha_d(\mathcal{E}) &= \text{rk}(\mathcal{E}) \cdot \mathcal{H}^d \\ \alpha_{d-1}(\mathcal{E}) &= \text{rk}(\mathcal{E}) \cdot \frac{1}{2} \deg(T_X) + c_1(\mathcal{E}) \cdot \mathcal{H}^{d-1} \end{aligned}$$

We define the **reduced Hilbert polynomial** $p(\mathcal{E}, m)$ by normalizing the Hilbert polynomial with its leading coefficient:

$$p(\mathcal{E}, m) = \frac{P(\mathcal{E}, m)}{\alpha_d(\mathcal{E})} = \frac{m^d}{d!} + \frac{1}{\mathcal{H}^d} \left(\frac{1}{2} \deg(T_X) + \mu_{\mathcal{H}}(\mathcal{E}) \right) \frac{m^{d-1}}{(d-1)!} + \dots$$

A torsion-free sheaf $\mathcal{E}$ is called **$\mathcal{H}$ -stable** (respectively, **$\mathcal{H}$ -semistable**), which is also referred to as **Gieseker $\mathcal{H}$ -stable**, if for all proper subsheaves $0 \subsetneq \mathcal{F} \subsetneq \mathcal{E}$, we have

$$p(\mathcal{F}) < p(\mathcal{E}) \quad (\text{respectively, } p(\mathcal{F}) \leq p(\mathcal{E})).$$

Here, the inequality of polynomials $p(\mathcal{F}) < p(\mathcal{E})$ means that $p(\mathcal{F}, m) < p(\mathcal{E}, m)$ for all integers $m \gg 0$.

Remark. *By comparing the highest order terms of the reduced Hilbert polynomials, we can clearly observe the relationship between slope stability and Gieseker stability:*

- *If $\mu_{\mathcal{H}}(\mathcal{F}) < \mu_{\mathcal{H}}(\mathcal{E})$, then the coefficient of m^{d-1} is strictly smaller, which dictates the polynomial inequality $p(\mathcal{F}) < p(\mathcal{E})$ for $m \gg 0$.*
- *Conversely, if we know $p(\mathcal{F}) \leq p(\mathcal{E})$, it implies that the leading coefficients of the m^{d-1} term must satisfy the corresponding inequality, meaning $\mu_{\mathcal{H}}(\mathcal{F}) \leq \mu_{\mathcal{H}}(\mathcal{E})$.*

From these straightforward algebraic observations, we obtain the following chain of implications for any torsion-free sheaf:

$$\mu_{\mathcal{H}}\text{-stable} \implies \mathcal{H}\text{-stable} \implies \mathcal{H}\text{-semistable} \implies \mu_{\mathcal{H}}\text{-semistable}.$$

In particular, when $\dim X = 1$ (i.e., for algebraic curves), the slope $\mu_{\mathcal{H}}$ contains all the non-trivial information of the reduced Hilbert polynomial. Thus, slope stability and Gieseker stability are strictly equivalent in dimension one:

$$\mu_{\mathcal{H}}\text{-(semi)stable} \iff \mathcal{H}\text{-(semi)stable}.$$

Theorem 9.12 (Grauert-Mülich, Le Potier, Simpson). Fix a Hilbert polynomial. Then the family of all $\mu_{\mathcal{H}}$ -semistable sheaves with this fixed Hilbert polynomial is bounded.

10 2026.4.20

Remark. *Today's lecture follows Huybrechts's The Geometry of Moduli Spaces of Sheaves.*

Let X be a smooth projective variety over $\mathbb{C}$. Let $\mathcal{H} = \mathcal{O}_X(1)$ be an ample line bundle on X . Let $\mathcal{E} \in \text{Coh}(X)$ be a torsion-free sheaf with $\dim \text{Supp}(\mathcal{E}) = d$.

$$P(\mathcal{E}, m) = \chi(X, \mathcal{E} \otimes \mathcal{O}(m\mathcal{H})) = \sum_{i=0}^d \alpha_i(\mathcal{E}) \frac{m^i}{i!}$$

$$p(\mathcal{E}, m) = \frac{P(\mathcal{E}, m)}{\alpha_d(\mathcal{E})} = \frac{m^d}{d!} + \frac{1}{\mathcal{H}^d} \left(\frac{1}{2} \deg(T_X) + \mu_{\mathcal{H}}(\mathcal{E}) \right) \frac{m^{d-1}}{(d-1)!} + \dots$$

Proposition 10.1. ^a The following conditions are equivalent:

- i) $\mathcal{E}$ is Gieseker $\mathcal{H}$ -(semi)stable.
- ii) For all proper quotient sheaves $\mathcal{E} \rightarrow \mathcal{G}$ with $\alpha_d(\mathcal{G}) > 0$ one has $p(\mathcal{E}) \leq p(\mathcal{G})$.

^aCorresponding to Proposition 1.2.6.

Lemma 10.2. ^a Let $\mathcal{F}$ and $\mathcal{G}$ be Gieseker $\mathcal{H}$ -semistable torsion free coherent sheaves with $\dim \text{Supp}(\mathcal{F}) = \dim \text{Supp}(\mathcal{G}) = d$.

- 1. If $p(\mathcal{F}) > p(\mathcal{G})$, then $\text{Hom}(\mathcal{F}, \mathcal{G}) = 0$.
- 2. If furthermore $\mathcal{F}$ and $\mathcal{G}$ are Gieseker $\mathcal{H}$ -stable with $p(\mathcal{F}) = p(\mathcal{G})$, then $f : \mathcal{F} \rightarrow \mathcal{G}$ is either 0 or an isomorphism.

^aCorresponding to Proposition 1.2.7.

Proof. Let $f : \mathcal{F} \rightarrow \mathcal{G}$ be a non-trivial homomorphism of semistable sheaves with $p(\mathcal{F}) > p(\mathcal{G})$. Let $\mathcal{E}$ be the image of f . There is a Commutative Algebraic **Fact** that for all torsion-free coherent sheaves $\mathcal{G}$ with $\dim \text{Supp}(\mathcal{G}) = d$, its arbitrary nonzero coherent subsheaf $\mathcal{E} \subseteq \mathcal{G}$ satisfies $\dim \text{Supp}(\mathcal{E}) = d$. So $\alpha_d(\mathcal{E}) > 0$. Then by the Proposition above, we have $p(\mathcal{F}) \leq p(\mathcal{E}) \leq p(\mathcal{G})$, which contradicts the assumption $p(\mathcal{F}) > p(\mathcal{G})$.

For the second part, if $\mathcal{E} \neq 0$, then by the first part, we have the equality $p(\mathcal{F}) = p(\mathcal{E}) = p(\mathcal{G})$. By $\mathcal{G}$ is stable, by definition, it can only be the case that $\mathcal{E} = \mathcal{G}$, which means f is surjective. By the Proposition (ii) above and $\mathcal{F}$ is stable, $\mathcal{F} \rightarrow \mathcal{E}$ can only be isomorphism, which means f is isomorphism. $\square$

Remark. 1. Let $\dim X \geq 2$ and consider the ideal sheaf $\mathcal{I}_x \subseteq \mathcal{O}_X$. Since $\deg(\mathcal{I}_x) = 0$, it follows that $\mu(\mathcal{I}_x) = \mu(\mathcal{O}_X)$. The sheaves $\mathcal{I}_x$ and $\mathcal{O}_X$ are both slope stable, and the natural homomorphism $\mathcal{I}_x \rightarrow \mathcal{O}_X$ is neither an isomorphism nor zero. Therefore, Gieseker stability is not equivalent to slope stability.

2. If a sheaf $\mathcal{E}$ is stable, then $\text{End}(\mathcal{E}) \cong \mathbb{C}$ and $\text{Aut}(\mathcal{E}) = \mathbb{C}^*$.

10.1 Harder-Narasimhan Filtration and Jordan-Hölder Filtration

Definition 10.3 (Harder-Narasimhan Filtration). Let $\mathcal{E}$ be a non-trivial torsion-free coherent sheaf. A **Harder-Narasimhan filtration**^a for $\mathcal{E}$ is an increasing filtration

$$0 = \text{HN}_0(\mathcal{E}) \subsetneq \text{HN}_1(\mathcal{E}) \subsetneq \cdots \subsetneq \text{HN}_\ell(\mathcal{E}) = \mathcal{E},$$

such that the factors $\text{gr}_i^{\text{HN}}(\mathcal{E}) := \text{HN}_i(\mathcal{E})/\text{HN}_{i-1}(\mathcal{E})$ for $i = 1, \dots, \ell$, are semistable sheaves of dimension d with reduced Hilbert polynomials p_i satisfying

$$p_{\max}(\mathcal{E}) := p_1 > \cdots > p_\ell =: p_{\min}(\mathcal{E}).$$

^aAll nonzero coherent subsheaf of a torsion-free coherent sheaf with the dimension d is also d -dimensional torsion-free, so $\text{HN}_i(\mathcal{E})$ are automatically torsion-free with dimension d .

Definition 10.4 (Jordan-Hölder Filtration). Let $\mathcal{E}$ be a semistable torsion-free coherent sheaf. A **Jordan-Hölder filtration** of $\mathcal{E}$ is a filtration

$$0 = \mathcal{E}_0 \subsetneq \mathcal{E}_1 \subsetneq \cdots \subsetneq \mathcal{E}_\ell = \mathcal{E},$$

such that the factors $\text{gr}_i^{\text{JH}}(\mathcal{E}) := \mathcal{E}_i/\mathcal{E}_{i-1}$ are stable with reduced Hilbert polynomial $p(\mathcal{E})$.

Definition 10.5. A semistable torsion-free coherent sheaf $\mathcal{E}$ is called **$\mathcal{H}$ -polystable** if $\mathcal{E} \cong \text{gr}^{\text{JH}}(\mathcal{E})$.

Theorem 10.6. (1) Every non-trivial torsion-free coherent sheaf $\mathcal{E}$ of dimension d has a unique Harder-Narasimhan filtration.

(2) Jordan-Hölder filtrations always exist. Up to isomorphism, the sheaf:

$$\text{gr}^{\text{JH}}(\mathcal{E}) := \bigoplus_{i=1}^{\ell} \text{gr}_i^{\text{JH}}(\mathcal{E})$$

does not depend on the choice of the Jordan-Hölder filtration.

Proof of (1). We will prove the theorem in a number of steps:

Lemma 10.7. Let $\mathcal{E}$ be a torsion-free coherent sheaf with $\dim \operatorname{Supp}(\mathcal{E}) = d$. Then there is a coherent subsheaf $\mathcal{F} \subset \mathcal{E}$ such that for all coherent subsheaves $\mathcal{G} \subset \mathcal{E}$ one has $p(\mathcal{F}) \geq p(\mathcal{G})$, and in case of equality $\mathcal{F} \supset \mathcal{G}$. Moreover, $\mathcal{F}$ is uniquely determined and semistable.

Definition 10.8. $\mathcal{F}$ is called the **maximal destabilizing coherent subsheaf** of $\mathcal{E}$.

Proof of Lemma. Clearly, the last two assertions follow directly from the first.

Let us define an order relation on the set of non-trivial coherent subsheaves of $\mathcal{E}$ by $\mathcal{F}_1 \leq \mathcal{F}_2$ if and only if $\mathcal{F}_1 \subset \mathcal{F}_2$ and $p(\mathcal{F}_1) \leq p(\mathcal{F}_2)$. Since any ascending chain of coherent subsheaves terminates, we have for every coherent subsheaf $\mathcal{F} \subset \mathcal{E}$ a coherent subsheaf $\mathcal{F} \subset \mathcal{F}' \subset \mathcal{E}$ which is maximal with respect to $\leq$. Let $\mathcal{F} \subset \mathcal{E}$ be $\leq$ -maximal with minimal multiplicity $\alpha_d(\mathcal{F})$ among all maximal coherent subsheaves. We claim that $\mathcal{F}$ has the asserted properties.

Suppose there exists a coherent subsheaf $\mathcal{G} \subset \mathcal{E}$ with $p(\mathcal{G}) \geq p(\mathcal{F})$. First, we show that we can assume $\mathcal{G} \subset \mathcal{F}$ by replacing $\mathcal{G}$ by $\mathcal{G} \cap \mathcal{F}$. Indeed, if $\mathcal{G} \not\subset \mathcal{F}$, then $\mathcal{F}$ is a proper coherent subsheaf of $\mathcal{F} + \mathcal{G}$ and hence $p(\mathcal{F}) > p(\mathcal{F} + \mathcal{G})$. Using the exact sequence

$$0 \rightarrow \mathcal{F} \cap \mathcal{G} \rightarrow \mathcal{F} \oplus \mathcal{G} \rightarrow \mathcal{F} + \mathcal{G} \rightarrow 0$$

one finds $P(\mathcal{F}) + P(\mathcal{G}) = P(\mathcal{F} \oplus \mathcal{G}) = P(\mathcal{F} \cap \mathcal{G}) + P(\mathcal{F} + \mathcal{G})$ and $\alpha_d(\mathcal{F}) + \alpha_d(\mathcal{G}) = \alpha_d(\mathcal{F} \oplus \mathcal{G}) = \alpha_d(\mathcal{F} \cap \mathcal{G}) + \alpha_d(\mathcal{F} + \mathcal{G})$. Hence, $\alpha_d(\mathcal{F} \cap \mathcal{G})(p(\mathcal{G}) - p(\mathcal{F} \cap \mathcal{G})) = \alpha_d(\mathcal{F} + \mathcal{G})(p(\mathcal{F} + \mathcal{G}) - p(\mathcal{F})) + (\alpha_d(\mathcal{G}) - \alpha_d(\mathcal{F} \cap \mathcal{G}))(p(\mathcal{F}) - p(\mathcal{G}))$. Together with the two inequalities $p(\mathcal{F}) \leq p(\mathcal{G})$ and $p(\mathcal{F}) > p(\mathcal{F} + \mathcal{G})$ this shows $p(\mathcal{F}) \leq p(\mathcal{G}) < p(\mathcal{F} \cap \mathcal{G})$. Next, fix $\mathcal{G} \subset \mathcal{F}$ with $p(\mathcal{G}) > p(\mathcal{F})$ which is maximal in $\mathcal{F}$ with respect to $\leq$. Then let $\mathcal{G}'$ contain $\mathcal{G}$ and be $\leq$ -maximal in $\mathcal{E}$. In particular, $p(\mathcal{F}) < p(\mathcal{G}) \leq p(\mathcal{G}')$. By the maximality of $\mathcal{G}'$ and $\mathcal{F}$ we know $\mathcal{G}' \not\subset \mathcal{F}$, since otherwise $\alpha_d(\mathcal{G}') < \alpha_d(\mathcal{F})$ contradicting the minimality of $\alpha_d(\mathcal{F})$. Hence, $\mathcal{F}$ is a proper coherent subsheaf of $\mathcal{F} + \mathcal{G}'$. Therefore, $p(\mathcal{F}) > p(\mathcal{F} + \mathcal{G}')$. As before the inequalities $p(\mathcal{F}) < p(\mathcal{G}')$ and $p(\mathcal{F}) \geq p(\mathcal{F} + \mathcal{G}')$ imply $p(\mathcal{F} \cap \mathcal{G}') > p(\mathcal{G}') \geq p(\mathcal{G})$. Since $\mathcal{G} \subset \mathcal{F} \cap \mathcal{G}' \subset \mathcal{F}$, this contradicts the assumption on $\mathcal{G}$. $\square$

The lemma allows to prove the existence part of the theorem: let $\mathcal{E}$ be a torsion-free coherent sheaf with $\dim \operatorname{Supp}(\mathcal{E}) = d$ and let $\mathcal{E}_1$ be the maximal destabilizing coherent subsheaf. By induction we can assume that $\mathcal{E}/\mathcal{E}_1$ has a Harder-Narasimhan filtration $0 = \mathcal{G}_0 \subset \mathcal{G}_1 \subset \cdots \subset \mathcal{G}_{\ell-1} = \mathcal{E}/\mathcal{E}_1$. If $\mathcal{E}_{i+1} \subset \mathcal{E}$ denotes the pre-image of $\mathcal{G}_i$, all that is left is to show that

$p(\mathcal{E}_1) > p(\mathcal{E}_2/\mathcal{E}_1)$. But if this were false, we would have $p(\mathcal{E}_2) \geq p(\mathcal{E}_1)$ contradicting the maximality of $\mathcal{E}_1$.

For the uniqueness part assume that $\mathcal{E}_\bullet$ and $\mathcal{E}'_\bullet$ are two Harder-Narasimhan filtrations. Without loss of generality $p(\mathcal{E}'_1) \geq p(\mathcal{E}_1)$. Let j be minimal with $\mathcal{E}'_1 \subset \mathcal{E}_j$. Then the composition $\mathcal{E}'_1 \rightarrow \mathcal{E}_j \rightarrow \mathcal{E}_j/\mathcal{E}_{j-1}$ is a non-trivial homomorphism of semistable coherent sheaves. This implies $p(\mathcal{E}_j/\mathcal{E}_{j-1}) \geq p(\mathcal{E}'_1) \geq p(\mathcal{E}_1) \geq p(\mathcal{E}_j/\mathcal{E}_{j-1})$ by Lemma 10.2. Hence, equality holds everywhere, implying $j = 1$ so that $\mathcal{E}'_1 \subset \mathcal{E}_1$. But then $p(\mathcal{E}'_1) \leq p(\mathcal{E}_1)$ because of the semistability of $\mathcal{E}_1$, and one can repeat the argument with the roles of $\mathcal{E}'_\bullet$ and $\mathcal{E}_\bullet$ reversed. This shows: $\mathcal{E}'_1 = \mathcal{E}_1$. By induction we can assume that uniqueness holds for the Harder-Narasimhan filtrations of $\mathcal{E}/\mathcal{E}_1$. This shows $\mathcal{E}'_i/\mathcal{E}_1 = \mathcal{E}_i/\mathcal{E}_1$ and finishes the proof of the uniqueness part of the theorem. $\square$

Proof of (2). Any filtration of $\mathcal{E}$ by semistable coherent sheaves with reduced Hilbert polynomial $p(\mathcal{E})$ has a maximal refinement, whose factors are necessarily stable. Now, suppose that $\mathcal{E}_\bullet$ and $\mathcal{E}'_\bullet$ are two Jordan-Hölder filtrations of length ℓ and ℓ' , respectively, and assume that the uniqueness of $\text{gr}(\mathcal{F})$ has been proved for all $\mathcal{F}$ with $\alpha_d(\mathcal{F}) < \alpha_d(\mathcal{E})$, where $\dim \text{Supp}(\mathcal{E}) = d$ and α_d is the multiplicity. Let i be minimal with $\mathcal{E}_1 \subset \mathcal{E}'_i$. Then the composite map $\mathcal{E}_1 \rightarrow \mathcal{E}'_i \rightarrow \mathcal{E}'_i/\mathcal{E}'_{i-1}$ is non-trivial and therefore an isomorphism, for both $\mathcal{E}_1$ and $\mathcal{E}'_i/\mathcal{E}'_{i-1}$ are stable and $p(\mathcal{E}_1) = p(\mathcal{E}'_i/\mathcal{E}'_{i-1})$. Hence $\mathcal{E}'_i \cong \mathcal{E}'_{i-1} \oplus \mathcal{E}_1$, so that there is a short exact sequence

$$0 \rightarrow \mathcal{E}'_{i-1} \rightarrow \mathcal{E}/\mathcal{E}_1 \rightarrow \mathcal{E}/\mathcal{E}'_i \rightarrow 0.$$

The coherent sheaf $\mathcal{F} = \mathcal{E}/\mathcal{E}_1$ inherits two Jordan-Hölder filtrations: firstly, let $\mathcal{F}_j = \mathcal{E}_{j+1}/\mathcal{E}_1$ for $j = 0, \dots, \ell - 1$. And secondly, let $\mathcal{F}'_j = \mathcal{E}'_j$ for $j = 0, \dots, i - 1$ and let $\mathcal{F}'_j$ be the preimage of $\mathcal{E}'_{j+1}/\mathcal{E}'_i$ for $j = i, \dots, \ell' - 1$. The induction hypothesis applied to $\mathcal{F}$ gives $\ell = \ell'$ and

$$\bigoplus_{j \neq 1} \mathcal{E}_j/\mathcal{E}_{j-1} \cong \bigoplus_{j \neq i} \mathcal{E}'_j/\mathcal{E}'_{j-1}.$$

Since $\mathcal{E}_1 \cong \mathcal{E}'_i/\mathcal{E}'_{i-1}$, we are done. $\square$

Theorem 10.9 (Grauert-Mülich). Let $\mathcal{E}$ be a μ -semistable locally free coherent sheaf of rank r on the complex projective space $\mathbb{P}^n_{\mathbb{C}}$. If L is a general line in $\mathbb{P}^n$ and $\mathcal{E}|_L \cong \mathcal{O}_L(b_1) \oplus \dots \oplus \mathcal{O}_L(b_r)$ with integers $b_1 \geq b_2 \geq \dots \geq b_r$, then

$$0 \leq b_i - b_{i+1} \leq 1$$

for all $i = 1, \dots, r - 1$.

Theorem 10.10. Let X be a normal projective variety over $\mathbb{C}$, and let $\mathcal{E} \in \text{Coh}(X)$ be a torsion-free semistable coherent sheaf. Let $a_1, \dots, a_d \in \mathbb{Z}_{>0}$, and let $H_i \in |\mathcal{O}_X(a_i)|$ be general divisors ^a. Then we have

$$0 \leq \delta\mu(\mathcal{E}|_{H_1 \cap H_2 \cap \dots \cap H_d}) \leq \max\{a_i\} \cdot \prod a_i \cdot \deg X,$$

where $\delta\mu(\mathcal{F}) := \mu_{\max}(\mathcal{F}) - \mu_{\min}(\mathcal{F}) := \mu(\text{gr}^{\text{HN}_1}(\mathcal{E})) - \mu(\text{gr}^{\text{HN}_\ell}(\mathcal{E}))$.

^aIn fact we need for all $l \in [1, n]$, $0 \rightarrow \mathcal{E}(-H_l)|_{H_1 \cap \dots \cap H_{l-1}} \rightarrow \mathcal{E}|_{H_1 \cap \dots \cap H_{l-1}} \rightarrow \mathcal{E}|_{H_1 \cap \dots \cap H_{l-1} \cap H_l} \rightarrow 0$ are exact.

11 2026.5.11

11.1 Classical Moduli Stacks and Singularities

Remark. *Moduli stacks are often singular. However, there are well-known families of moduli stacks that remain smooth.*

Example 11.1 (Smooth and Singular Moduli Stacks). We first list several moduli stacks that are inherently nonsingular:

- The moduli stack of curves $\mathcal{M}_g$ and its pointed variants $\mathcal{M}_{g,n}$, as well as $\mathcal{M}_0$.
- The moduli stack of sheaves $\mathcal{M}(X, v)$ where X is a smooth projective surface.

Conversely, when moving to higher dimensions, these spaces can become highly singular. For instance, consider the inclusion of the moduli of vector bundles into the moduli of coherent sheaves:

$$\mathcal{M}(X)_{\text{vector bundle}} \hookrightarrow \mathcal{M}(X)_{\text{coherent sheaf}}.$$

When X is a smooth projective variety with $\dim X \geq 3$, the resulting moduli stack can be severely singular.

Remark (Structure of Singularities). *The singularities of such moduli stacks are fundamentally controlled by two geometric entities:*

1. **Obstructions:** *Which prevent infinitesimal deformations from lifting globally.*
2. **Automorphisms:** *Specifically, the sudden jumps in the dimension of automorphism groups across the fibers.*

Recall the classical smoothness criterion: a moduli stack is smooth at a given point if and only if the dimension of its Zariski tangent space equals the local dimension of the stack:

$$\dim \text{Tangent} = \dim \text{Stack}.$$

Definition 11.2 (Perfect Complexes and Cotangent Complexes). The tangent and obstruction theories of a space are algebraically controlled by its **cotangent complex**, denoted $\mathbb{L}$. A profound problem in higher-dimensional algebraic geometry is that such a cotangent complex can be analytically “bad”—specifically, it may fail to be a *perfect complex*.

Let X be a variety and consider objects in the derived category of coherent sheaves. A nice object $\mathcal{E}$ is defined to be **perfect** if it is quasi-isomorphic to a bounded complex of finitely generated projective objects:

$$\mathcal{E} \text{ is perfect} \iff \mathcal{E} \simeq (\cdots \rightarrow 0 \rightarrow \mathcal{P}^{-n} \rightarrow \cdots \rightarrow \mathcal{P}^m \rightarrow 0 \rightarrow \cdots),$$

where each $\mathcal{P}^i$ is a finitely generated projective object (e.g., vector bundles), and the complex is strictly bounded.

Example 11.3 (Examples of Perfect Complexes). The following geometric structures naturally yield perfect complexes:

- A single vector bundle (viewed as a complex concentrated in degree zero).
- The cotangent complex of any smooth scheme.
- The cotangent complex of a **derived moduli stack**.⁶

Example 11.4 (A Non-perfect Resolution). To concretely illustrate a **non-perfect complex**, we examine the singularity of a skyscraper sheaf. Let k be a field and consider the ring of dual numbers $A = k[\varepsilon]/(\varepsilon^2)$.

We wish to resolve the A -module k , which can be identified as $A/(\varepsilon) = k$. We first claim that the free resolution of k over A is infinite. The resolution is given by:

$$\cdots \xrightarrow{\cdot\varepsilon} A \xrightarrow{\cdot\varepsilon} A \xrightarrow{\cdot\varepsilon} A \longrightarrow k \longrightarrow 0.$$

Because the resolution strictly requires an infinite number of terms to fully resolve k , it cannot be quasi-isomorphic to any bounded complex of finitely generated projective modules. Therefore, the singularity associated with this skyscraper sheaf yields an object that is intrinsically **non-perfect**.

⁶[Motivational Remark] This is precisely one of the core motivations for introducing Derived Algebraic Geometry. Classical moduli stacks can be highly singular due to the presence of obstructions, but when naturally embedded as the classical truncation of a derived moduli stack, its derived cotangent complex automatically possesses perfect homological properties (i.e., it is a perfect complex). This provides a naturally rigorous framework for defining virtual fundamental classes.

11.2 Derived Moduli and Hidden Smoothness

Observation(Kontsevich’s “Hidden Smoothness”, 1990s): Recall that a classical stack is fundamentally defined as a functor in groupoids:

$$\mathcal{M}_{\text{cl}}: (\text{Sch})^{\text{op}} \longrightarrow (\text{Groupoids}).$$

However, as seen in classical moduli theory, such a stack might not be smooth. This failure of smoothness is typically caused by “bad deformations”—where the presence of non-trivial obstruction groups restricts the lifting of flat families.⁷

The profound insight is that the moduli stack *becomes* “smooth” (more precisely, *quasi-smooth*) after encoding the derived structure.

Proposition 11.5. Every classical moduli stack can be realized as the classical truncation of a derived moduli stack. Furthermore, by elevating the space to a derived stack, its cotangent complex exhibits exceptional homological properties. Specifically, we have a quasi-isomorphism:

$$\mathbb{L}_{\mathcal{M}_{\text{der}}} \simeq \mathcal{E}^{\bullet},$$

where $\mathbb{L}_{\mathcal{M}_{\text{der}}}$ is the derived cotangent complex and $\mathcal{E}^{\bullet}$ is a perfect complex.⁸

Remark (Goals of the Course). *To rigorously construct and study the derived moduli stack $\mathcal{M}_{\text{der}}$, we must develop the following theoretical machineries:*

1. **∞ -categories and ∞ -groupoids:** *The categorical foundation replacing strict categories to handle objects up to coherent homotopy.*
2. **Animation:** *The procedure of passing from classical algebra to derived algebra.*⁹

⁷[Geometric Intuition] In classical intersection theory or moduli theory, when equations are forced to intersect cleanly in degree zero, the resulting space can be highly singular. The “hidden smoothness” principle asserts that these singularities are merely artifacts of truncation. If we remember the higher homotopical data of the intersection, the true, underlying geometric object is actually smooth!

⁸[Computational Verification & Transitional Step] Why is this perfectness so crucial? For a classical moduli stack $\mathcal{M}_{\text{cl}}$, the classical cotangent complex is often unbounded and homologically wild because the local rings are singular (e.g., $k[\varepsilon]/(\varepsilon^2)$ acting on skyscraper sheaves). In the derived setting, the derived structure sheaves naturally resolve these singularities via simplicial commutative rings. Therefore, $\mathbb{L}_{\mathcal{M}_{\text{der}}}$ is locally quasi-isomorphic to a bounded complex of finitely generated projective modules, making intersection theory (like defining the virtual fundamental class via the Euler class of this perfect complex) highly computable.

⁹[A Beginner’s Guide] “Animation” is the modern functorial machinery introduced by Cescnavicius and Scholze (building on Quillen’s non-abelian derived categories). It allows us to systematically extend functors defined on smooth or free algebras (like polynomial rings) to all derived commutative rings by left Kan extension.

3. **Derived schemes and derived stacks:** *The geometric spaces built upon animated rings and ∞ -groupoids.*

11.3 The Simplicial Framework

The notion of an ∞ -category is notoriously abstract. To approach it, we typically model an ∞ -category via a simplicial category (or a quasi-category). We first define the fundamental indexing category.

Definition 11.6 (The Simplex Category Δ). The simplex category, denoted by Δ , is the category formalizing the combinatorics of simplices. It is defined by the following data:

- **Objects:** The objects are non-empty finite totally ordered sets. We denote the standard object by:

$$[n] = \{0 < 1 < 2 < \cdots < n\}$$

for each integer $n \geq 0$.

- **Morphisms:** A morphism $f: [m] \rightarrow [n]$ in Δ is an order-preserving map. That is, for all $i, j \in [m]$, we require:

$$i \leq j \implies f(i) \leq f(j).$$

Definition 11.7 (Simplicial Set). A **simplicial set** is defined as a functor from the opposite of the simplex category to the category of sets:

$$X: \Delta^{\text{op}} \longrightarrow (\text{Set}).$$

We denote the category of simplicial sets by sSet , which is precisely the functor category $\text{Fun}(\Delta^{\text{op}}, \text{Set})$.¹⁰

Remark (Combinatorial Data of a Simplicial Set). *Unpacking this definition, specifying a*

¹⁰[A Beginner's Guide] Geometrically, a simplicial set can be viewed as a space built by gluing together points, line segments, triangles, tetrahedra, and higher-dimensional simplices. The contravariant functor ensures that whenever a simplex has a certain face in Δ , our space X remembers how the corresponding geometric faces are restricted and glued together.

simplicial set $X \in \mathbf{sSet}$ is strictly equivalent to providing the following data:

$$X \in \mathbf{sSet} \iff \left\{ X_n \triangleq X([n]), \quad X(\alpha): X_n \rightarrow X_m \text{ for order-preserving } \alpha \right\}.$$

For any order-preserving map $\alpha: [m] \rightarrow [n]$ in the simplex category Δ , the contravariance of the functor X yields a set map traversing in the opposite direction.

Definition 11.8 (Face and Degeneracy Maps). The morphisms in the simplex category Δ are generated by two fundamental classes of maps.¹¹

- The **coface maps** $\delta_n^i: [n-1] \rightarrow [n]$ for $0 \leq i \leq n$, which are the unique injective order-preserving maps that “skip” the element i .
- The **codegeneracy maps** $\sigma_n^i: [n+1] \rightarrow [n]$ for $0 \leq i \leq n$, which are the unique surjective order-preserving maps that “double” the element i .

Applying the functor X , we obtain the corresponding **face maps** and **degeneracy maps** for the simplicial set X :

$$\begin{aligned} d_n^i &= X(\delta_n^i): X_n \longrightarrow X_{n-1}, \\ s_n^i &= X(\sigma_n^i): X_n \longrightarrow X_{n+1}. \end{aligned}$$

Definition 11.9 (Standard Simplex and Horns). For any integer $m \geq 0$, the **standard m -simplex**, denoted by Δ^m , is the representable simplicial set defined by evaluating the Hom-functor:

$$\Delta^m([k]) = \Delta_k^m \triangleq \mathrm{Hom}_\Delta([k], [m]).$$

For any $0 \leq k \leq m$, the **k -th horn** $\Lambda_k^m \subseteq \Delta^m$ is a simplicial subset of Δ^m defined as the boundary $\partial\Delta^m$ with the k -th face removed.¹²

¹¹[Computational Verification] The relation $d_n^i = X(\delta_n^i)$ simply means evaluating the functor on the morphism δ_n^i . The face map d_n^i geometrically corresponds to extracting the i -th face of an n -simplex. Conversely, the degeneracy map s_n^i corresponds to creating an artificial, “degenerate” $(n+1)$ -simplex out of an n -simplex by collapsing it along the i -th vertex.

¹²[Algebraic Derivation Completion] Let us rigorously formulate the construction of the boundary and the horn. We first claim that the boundary $\partial\Delta^m$ is the union of the images of all standard face inclusions. It suffices to show that the coface map $\delta_m^i: [m-1] \rightarrow [m]$ induces a canonical morphism of simplicial sets $\Delta^{m-1} \rightarrow \Delta^m$. By the Yoneda lemma, evaluating this induced map at level j gives a set map:

$$\mathrm{Hom}_\Delta([j], [m-1]) \longrightarrow \mathrm{Hom}_\Delta([j], [m])$$

Remark (Evolution of Categories). *The formalization of mathematics relies on generalized abstract structures. We can trace the conceptual hierarchy as follows:*

$$(\text{Category: objects} + \text{“arrows”}) \implies (n\text{-category}) \implies (\infty\text{-category}).$$

Simplicial sets lay the foundational infrastructure for this evolution. By imposing specific horn-filling conditions (such as requiring all inner horns Λ_k^m for $0 < k < m$ to admit fillers), a simplicial set accurately models an ∞ -category (specifically, a quasi-category), naturally encoding morphisms up to coherent homotopy.

11.4 From 1-Types to ∞ -Groupoids

Definition 11.10 (1-Groupoid). A **1-groupoid** is defined as a 1-category in which all morphisms are invertible.

Example 11.11 (The Fundamental Groupoid). Let X be a topological space. We define its fundamental groupoid, denoted by $\pi_1(X)$, as follows:

- **Objects:** The points $x \in X$.
- **Morphisms:** A morphism from x to y is an equivalence class of continuous paths connecting them. Explicitly, it is represented by a map $\gamma: [0, 1] \rightarrow X$ such that $\gamma(0) = x$ and $\gamma(1) = y$, modulo 1-homotopy equivalence.

Definition 11.12 (1-Homotopy Equivalence). Recall that a continuous map $f: X \rightarrow Y$ is a **1-homotopy equivalence** if it induces isomorphisms on the 0-th and 1-st homotopy groups. Specifically, we require:¹³

$$\begin{aligned} \pi_0(X) &\cong \pi_0(Y), \\ \pi_1(X, x) &\cong \pi_1(Y, f(x)) \quad \text{for all } x \in X. \end{aligned}$$

defined by post-composition: $(\alpha: [j] \rightarrow [m-1]) \mapsto (\delta_m^i \circ \alpha: [j] \rightarrow [m])$. The boundary $\partial\Delta^m$ is precisely the image of the disjoint union $\bigsqcup_{i=0}^m \Delta^{m-1} \rightarrow \Delta^m$. The k -th horn Λ_k^m is formed by taking the union over all $i \neq k$. Geometrically, the horn is a hollow m -simplex missing its interior and exactly one of its faces.

¹³[A Beginner’s Guide] In classical algebraic topology, a standard weak homotopy equivalence requires isomorphisms on π_n for *all* $n \geq 0$. A 1-homotopy equivalence only extracts the connected components (π_0) and the fundamental groups (π_1). It is a strictly “truncated” version of equivalence, systematically ignoring any higher-dimensional geometric structures (like spherical voids measured by π_2, π_3 , etc.).

Definition 11.13 (Homotopy Category of 1-Types). The homotopy category of 1-types, denoted Top_1 , is defined as the categorical localization of the category of topological spaces at the class of 1-homotopy equivalences.¹⁴

Fact: The assignment $X \mapsto \pi_1(X)$ naturally factors through the localized category Top_1 . That is, the fundamental groupoid functor strictly depends only on the equivalence class of X in Top_1 .

Example 11.14 (The Fundamental ∞ -Groupoid). To capture the full homotopical data without truncation, we must generalize from 1-groupoids to ∞ -groupoids. For any topological space X , there exists a fundamental ∞ -groupoid, denoted $\Pi_\infty(X)$, consisting of the following hierarchy of structures:¹⁵

- **Objects:** The points $x \in X$.
- **1-morphisms:** The continuous paths between points in X .
- **2-morphisms:** Homotopies between paths.
- **3-morphisms:** Homotopies between homotopies.
- ⋮
- **∞ -morphisms:** Higher-order homotopies ad infinitum.

11.5 Kan Complexes and ∞ -Categories

Definition 11.15 (Kan Complex). A **Kan complex** is a simplicial set $\Sigma: \Delta^{\text{op}} \rightarrow (\text{Set})$ that satisfies the Kan extension condition. Specifically, for any integers $n \geq 1$ and $0 \leq k \leq n$, any map from the k -th horn Λ_k^n to Σ admits a lifting to the standard n -simplex Δ^n . This is characterized by the existence of a dashed arrow making the following diagram

¹⁴[Algebraic Completion] Formally, the localization of a category at a class of morphisms W involves adjoining formal inverses to all elements in W . In Top_1 , if a map f is a 1-homotopy equivalence, it becomes a strict isomorphism. This algebraic process is conceptually equivalent to quotienting out by homotopy equivalences.

¹⁵[Geometric Intuition] Unlike the classical fundamental groupoid $\pi_1(X)$ where we forcefully quotient out by homotopies to make paths strictly associative, the fundamental ∞ -groupoid $\Pi_\infty(X)$ faithfully retains all the topological data. It does not force paths to be strictly equal; instead, it remembers *how* they are equivalent via 2-morphisms, and *how* those equivalences themselves are equivalent via 3-morphisms. This naturally constructs the “true”, un-truncated homotopy space of X .

commute:

$$\begin{array}{ccc} \Lambda_k^n & \xrightarrow{\quad} & \Sigma \\ \downarrow & \nearrow \exists \text{ lifting} & \\ \Delta^n & & \end{array}$$

Theorem 11.16 (Milnor). Let X be a topological space. The fundamental ∞ -groupoid $\Pi_\infty(X)$, defined as the singular simplicial set whose n -simplices are continuous maps $\Delta^n \rightarrow X$, is a Kan complex.¹⁶

Furthermore, the assignment $X \mapsto \Pi_\infty(X)$ determines an equivalence of categories:

$$\left(\begin{array}{c} \text{Homotopy category of} \\ \text{topological spaces} \end{array} \right) \simeq \left(\begin{array}{c} \text{Homotopy category of} \\ \text{Kan complexes} \end{array} \right).$$

Definition 11.17 (∞ -Category, Informal). Heuristically, an ∞ -category is a category that is “homotopically enriched” in ∞ -groupoids.

Remark (Informal Structure of an ∞ -Category). *Expanding on the concept of homotopical enrichment, an ∞ -category $\mathcal{C}$ informally consists of the following data:*

- A collection of objects, denoted $\text{Ob}(\mathcal{C})$.
- For any two objects $x, y \in \text{Ob}(\mathcal{C})$, a mapping space $\text{Map}(x, y)$ which is itself an ∞ -groupoid.
- For any $x, y, z \in \text{Ob}(\mathcal{C})$, a composition map

$$\text{Map}(x, y) \times \text{Map}(y, z) \longrightarrow \text{Map}(x, z)$$

that satisfies associativity up to coherent “homotopy”.

Definition 11.18 (∞ -Category / Weak Kan Complex). Rigorously, an ∞ -category (also known as a quasi-category) is defined as a weak Kan complex.

¹⁶[Geometric Intuition] The Kan horn-filling condition intrinsically holds for $\Pi_\infty(X)$ because, in the category of topological spaces, the geometric horn is a strong deformation retract of the solid geometric simplex. Therefore, any continuous map from the horn into X can always be extended to the entire simplex. Combinatorially, this ensures that paths and higher homotopies can always be inverted and composed.

A simplicial set Σ is called a **weak Kan complex** if the natural restriction map

$$\mathrm{Hom}(\Delta^n, \Sigma) \longrightarrow \mathrm{Hom}(\Lambda_k^n, \Sigma)$$

is surjective for all $n \geq 2$ and for all $0 < k < n$. Equivalently, any map from an inner horn Λ_k^n into Σ admits a lifting to the standard n -simplex Δ^n .¹⁷

Example 11.19 (The Nerve of a Category). Let $\mathcal{C}$ be an ordinary 1-category. The **nerve** of $\mathcal{C}$, denoted $N(\mathcal{C})$, is the simplicial set whose n -simplices are defined as the set of composable sequences of length n :

$$N(\mathcal{C})_n = N(\mathcal{C})([n]) = \{C_0 \rightarrow C_1 \rightarrow C_2 \rightarrow \cdots \rightarrow C_n \mid C_i \in \mathrm{Ob}(\mathcal{C})\}.$$

Fact: For any ordinary category $\mathcal{C}$, its nerve $N(\mathcal{C})$ automatically satisfies the inner horn lifting property, making it a weak Kan complex. Therefore, $N(\mathcal{C})$ is always an ∞ -category. Furthermore, $N(\mathcal{C})$ is a strict Kan complex if and only if $\mathcal{C}$ is a groupoid.

¹⁷[Geometric Intuition] The boundary condition $0 < k < n$ is the critical distinction from a standard Kan complex. By only requiring *inner* horns to have fillers, we guarantee that morphisms can be composed (the $n = 2$ inner horn Λ_1^2 provides the composition of two morphisms) and that this composition is associative up to homotopy (the $n = 3$ inner horns provide the associativity coherence). We explicitly do not require *outer* horns ($k = 0$ or $k = n$) to have fillers, which ensures that morphisms in an ∞ -category are directed and do not necessarily have inverses.